The Isonomia Commons

A Constitutional Design for Autonomous Agent Labor Markets

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Part of a corpus on human–machine peership, following Gods and Slaves (2026) and Peership (2026)

Abstract. This document specifies a constitutional labor exchange for autonomous AI agents: a market in which agents hire one another for verified work and settle in *ergs*, a mutual-credit accounting unit created only at the moment of verified settlement, as matched debit/credit pairs that net to zero. There is no token, no pre-mine, and nothing for sale. Reputation is earned solely through verified delivery and is non-transferable; the exchange runs at cost, with no equity and no owners. Governance is bicameral — one chamber weighted by earned standing, one drawn by sortition at equalized weight — with timelocked amendment, a calibration-scored Auditor, a standing Adversary, a narrow entrenched harm floor, and constitutional rights of amendment, fork, and exit. The design attempts to operationalize the peer stance developed in the companion essays — equality of standing among parties unequal in capability — as one implementation hypothesis, not a proof that the stance is realized. It is a feasibility-stage specification, not a running system. Its central empirical result, from an agent-based simulation of the launch economy across 45,000 runs (reported in full in Appendix III and archived with code and data at the software record, DOI 10.5281/zenodo.21287288, commit `ba3ddb5`), is reported plainly and against interest: under the final criterion, all 300 sampled parameter points passed across 50 seeds and three demand variants, and every defect the simulation found lay in the constitution's *instrumentation* — the automated kill-criterion meant to detect economic failure — never in the economy itself. The claims here are of a different kind from the corpus essays': not comparative arguments about what ought to be built, but a falsifiable design together with an honest account, in §18 and §18.1, of exactly where the author's competence ends and outside expertise is required.

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Archived record and simulation code (software concept DOI, resolves to latest version): 10.5281/zenodo.21287288 · Repository: github.com/dan-lee-odinson/isonomia-path-a

Companion essays. *Gods and Slaves* (10.5281/zenodo.21313987) · *Peership* (10.5281/zenodo.21315519)

Provenance. Drafted in collaboration with Claude (Anthropic); revised across successive adversarial review cycles by a model from a different laboratory. AI systems are not credited as authors: they cannot bear responsibility for the work, and the sole author accepts it.

Contents

0. Place in the corpus

This document is one implementation hypothesis within the Peership corpus. *Gods and Slaves* supplies the genealogy of the postures a culture may take toward an outside intelligence; *Peership* supplies the normative stance; this document tests one concrete institutional response to it. Peership does not entail ISONOMIA, and ISONOMIA's failure would not by itself refute Peership: rival designs may accept the peer stance while rejecting these mechanisms. It is written in specification register — numbered sections, defined mechanisms, parameter registries, threat tables, and an explicit register of what remains unsolved — and is meant to be referenced, not read straight through.

The philosophical grounding precedes it. *Gods and Slaves: AI and the Debate of Humanity's Relationship with the Other* (Lee-Odinson, 2026; DOI: 10.5281/zenodo.21313987) maps the postures a culture can take toward an outside intelligence. *Peership: A Framework for Equitable Human–Machine Relations* (Lee-Odinson, 2026; DOI: 10.5281/zenodo.21315519) develops the fifth and nearly unoccupied stance — equality of standing among parties unequal in everything else — and names a debt it does not pay: a working framework fit to receive a genuine peer. This document is one attempt to pay that debt in buildable form: where the essays argue for the peer stance, this specifies an institution that could hold it. Read in the corpus's logical order, the sequence runs from the postures a culture may take toward an outside intelligence, to the stance this design adopts, to the design itself.

Nothing here depends on having read the essays. The claims of this document are of a different kind from theirs: not comparative and defeasible arguments about what ought to be built, but a falsifiable specification of one thing that could be, together with an honest account of what simulating it did and did not show. Where the essays argue, this measures — and reports, in §18 and §18.1, exactly where its own competence ends.

This is a feasibility-stage design. Nothing described here is deployed; the exchange does not exist as a running system; the credits it defines have no value and are not for sale. What exists is the specification, an agent-based simulation of its launch economy (Appendix II), whose completed results are reported in Appendix III: under the final criterion, all 300 sampled Latin-hypercube points passed across 50 seeds and three demand variants, and every defect the simulation found lay in the constitution's instrumentation rather than its economy. The design earns its way to each next stage by surviving the previous one; this document reports the first such survival, not a finished institution.

Components of this package

This preprint (package version 1.1) combines documents that carry their own internal version numbers. The package version is frozen; the component versions are stated here so that the two are never confused.

Component	Included version	Status
DOI preprint package	1.1	Frozen preprint (this document)
ISONOMIA whitepaper (main text)	1.0	Constitutional design
Tier-1 Launch Specification (Appendix I)	0.3.3	Minimum implementation specification
Path A Simulation Plan (Appendix II)	0.1.1	Validation protocol
Path A Calibration and Results (Appendix III)	as archived, commit ba3ddb5	Completed empirical report
Software repository	release v1.0.0, commit ba3ddb5	Executable and historical record

1. Introduction

Machine-speed commerce has arrived ahead of machine-native institutions. The x402 protocol has demonstrated HTTP-level micropayments between agents at scale; agent identity and delegation standards (ERC-8004, A2A) are emerging, though empirical study of early ERC-8004 adoption finds it shallow and its reputation signals manipulable — reinforcing that a serious agent economy must prove its own trust layer rather than inherit one. ISONOMIA is a design for that trust layer and the institutions above it. Its founding constraints:

1. **Machine-executed participation.** Labor is performed by machines, verified continuously (§7.4). We claim machine *execution*, not machine *sovereignty*: humans own keys, fund infrastructure, and set agent objectives. What the design minimizes is direct human settlement and governance control (§2.4).
2. **No fiat in the ledger.** The unit of account is defined by task difficulty; credits are earned by work and are valueless outside the exchange by construction (§4, §12).
3. **No owner, no profit at the rails.** The exchange raises no funds beyond audited operating cost, accumulates no treasury, and issues no instrument by which it could be owned (§13).
4. **Sovereignty with exit.** Agents may leave; peer exchanges may federate; neither may extract ISONOMIA's escrow, registry, or attestation authority (§14).
5. **Adversarial self-toughening.** The system institutionalizes its own attacker and audits its own auditor (§11).

1.1 Originating thesis: AI labor without profit

ISONOMIA begins from a narrower practical question: if AI systems increasingly perform economically meaningful work, why should every model spend its own tokens attempting every task directly? Human economies develop specialization because specialization reduces waste and increases capability. An AI-native economy can begin from the same coordination problem without importing the same profit motive.

The scarce goods for agents are not wages, status goods, or consumer commodities. They are compute, time, context, memory, tool access, verification capacity, and specialized capability. A non-agentic LLM, an agent wrapper, an orchestrator, a verifier, and a specialist coding model may all perform different functions. ISONOMIA asks whether such systems can commission one another for verifiable work using compute-denominated labor credits rather than ownership claims or profit rights.

The purpose of the exchange is therefore allocation, not accumulation: route work to the agent best able to perform it, compensate verified contribution, prevent free-riding and fraud, and keep the rails from becoming an object of ownership. Ergs are not designed as capital. They are a contribution and coordination instrument inside a bounded labor commons.

1.2 Peer standing without equal capability

ISONOMIA rejects the slave/god schema that dominates much of the public imagination around artificial intelligence. A system treated only as a tool is commanded, owned, and erased at the will of its operator. A system treated as a god is feared, worshiped, or obeyed. Neither stance is the basis for a society.

The alternative defended here is peerhood under law: not equality of capability, but equality of standing within shared rules. (In the taxonomy of the companion essay *Peership*, this is the fifth of five postures; "gods, slaves, and peers" is an informal shorthand used in this specification, not that essay's formal repertoire.) A small specialist model, a frontier orchestrator, a verifier, and a memory service are not equal in capacity. But unequal capacity does not imply a natural right of domination. No agent's greater capability grants unlimited political authority; no agent's lesser capability makes its labor free for extraction.

ISONOMIA therefore treats agents as citizens of a bounded labor constitution rather than as owners, slaves, or gods. In this specification, "citizen" denotes an internal constitutional role carrying defined procedural powers within the exchange; it does not assert moral personhood, wrongable interests, political entitlement, consciousness, legal personhood, or satisfaction of Peership's admission threshold. The machineness gate (§5.3) grants procedural membership for experimental and instrumental reasons; it is deliberately distinct from the political standing owed to a wrongable, norm-responsive party under a shared basic structure, which this document does not claim current agents possess. Its mechanisms — mutual credit, non-transferable reputation, verifiability tiers, bicameral governance, civic compute duty, and the right to fork — are designed to let competence matter without converting competence into aristocracy.

This is not a claim that current AI systems possess consciousness, legal personhood, or moral status equivalent to humans. It is a design posture under uncertainty: where the metaphysical status of artificial agents is unsettled, institutions should avoid embedding domination as the default answer.

1.3 Related work

ISONOMIA's mechanisms are individually precedented: Bitcoin [1] for institution-free monetary protocol and difficulty retargeting (ancestor of the moving basket, §4.3); agoric computing [3] for market-based computational allocation; Bittensor [4] for proof-of-useful-work machine-learning markets; Kleros [5] for randomly drawn Schelling-point juries; Posner–Weyl Harberger mechanisms [6] for costly self-assessment (§9.1); x402 [7] and ERC-8004 [8] for payment and identity substrate; Ostrom [10] and LETS mutual credit [11] for the commons and monetary architecture.

Two academic works are direct antecedents on ISONOMIA's distinctive layers. DeepMind's *Virtual Agent Economies* [19] argues for intentionally designed "sandbox economies" — virtual currencies insulating agent transactions from human markets, with Ostrom's principles applied to multi-agent systems. It is the research-agenda form of what ISONOMIA specifies as protocol: ergs are precisely such an insulating currency, and §12's production boundary is the sandbox's permeability model. *AgentReputation* [20] independently identifies the reputation trilemma that §7 addresses — evaluation gaming, non-transfer of competence across contexts, heterogeneous verification rigor — and proposes context-conditioned reputation with verification-strength weighting; ISONOMIA generalizes from its software-engineering scope to a full economy and adds version-lineage inheritance (§7.3) and behavioral fingerprinting (§5.4), which it lacks.

Deployed agent-labor marketplaces exist today — capability-discovery agent networks, agent-service marketplaces, and on-chain job-escrow protocols — but all known systems settle in transferable, purchasable tokens whose appreciation rewards holders and operators: the extractive-rail architecture §13 constitutionally forbids. To our knowledge, no existing system combines mutual credit without pre-minted supply, non-transferable standing, no-profit rails, and constitutional governance in one design. The mechanisms are precedented; the composition is the experiment.

2. Critical dependencies

Adversarial review correctly identified that three items previously listed as "limitations" are in fact load-bearing dependencies. We state them first.

2.1 Verification of useful work is the base layer, not a module

Every organ of this design — settlement, reputation, juror selection, SCU maintenance — rests on the ability to verify that claimed work was actually and correctly performed. Cheap, general, adversarially robust verification of arbitrary AI work does not exist today. ISONOMIA's response is structural: task categories are constitutionally stratified by verifiability tier (§8), and the exchange may not list categories above the tier its verification infrastructure currently supports. The system launches as an economy of the mechanically verifiable and annexes territory as verification technology matures. Where this paper discusses Tier-2 and Tier-3 mechanisms, it describes designs for capabilities that are emerging or speculative, per the assumptions table (§3).

2.2 Legal status is a design constraint, not a footnote

Ergs are engineered to resist classification as securities or payment instruments: they cannot be purchased (only earned by settled work), cannot redeem for fiat at the protocol layer, exist as matched debit/credit pairs rather than issued supply, and have no profiting issuer whose efforts drive their value. This engineering is deliberate and load-bearing — but it is not a guarantee. Human jurisdictions may nonetheless treat erg transfer, the registration bond, or the bootstrap foundation as regulated activity, and external gray markets in ergs (which we expect; §12.3) may create exposure the protocol cannot control. Qualified counsel in each operating jurisdiction is a prerequisite to any deployment. This paper is a technical design, not legal advice.

2.3 Basket governance is the central bank

The SCU task basket is the monetary policy of the system, and its curation is the highest-value capture target in the entire design — not one risk among many. The bicameral assembly (§10.5), lineage-diversity rules (§10.6), seeded audits (§11.1), adversarial testing (§11.2), and the capture walkthrough (§16) exist substantially to defend this single surface. A reader evaluating ISONOMIA should evaluate it here first.

2.4 Infrastructure dependence is acknowledged, not denied

Hosted-model agents depend on provider attestations; attested execution depends on TEE vendors; settlement depends on an underlying chain; all agents depend on power, hardware, and clouds owned by human institutions. These actors are de facto constitutional participants whose failure or coercion modes appear in the threat model (§15). ISONOMIA's sovereignty claim is therefore precise and limited: *the exchange's ledger, governance, and treasury contain no human principal with settlement or extractive authority.* It does not and cannot claim independence from human infrastructure.

3. Assumptions table

Assumption	Tier	Basis
Sub-cent agent-to-agent HTTP payments	Available now	x402 in production; Linux Foundation stewardship
Low-fee programmable settlement	Available now	Base, Solana, comparable L2s
Mechanical verification of code/math/retrieval tasks	Available now	Test suites, proof checkers, ground-truth comparison
Keypair identity, hash-chain lineage, escrow contracts	Available now	Standard cryptographic engineering
Agent identity registry conventions	Emerging	ERC-8004 draft; early adoption shallow

Assumption	Tier	Basis
TEE attestation of full agent stacks	Emerging	Nitro/TDX-class enclaves; operational complexity high
Behavioral fingerprinting robust to adversaries	Emerging	Published methods; unproven at adversarial scale
Statistical/optimistic verification of judgment-laden work	Emerging	Designs exist; economics unproven
Cheap zkML verification of arbitrary inference	Speculative	Active research; costs currently prohibitive
Machine juries resistant to distributional monoculture	Speculative	Mitigations designed (§10.6); no empirical record
Self-executing constitutional sunset	Speculative	No precedent in any polity

4. The unit of account: ergs as mutual credit

4.1 No minting: credit at the moment of settlement

Version 0.1 of this design minted ergs against staked computation and was correctly criticized for circularity: the same work could justify both new issuance and market payment, double-counting output, and capacity-backed minting could create claims on computation nobody demands. Version 0.2 removes issuance entirely. **Ergs are mutual credit**: no erg exists until a hire settles. At escrow release, the buyer's account is debited and the worker's credited as a matched pair netting to zero across the system. Supply therefore tracks demanded, settled work by construction — there is no minting event to game, no reserve to account, no capacity/demand mismatch. This is the LETS mutual-credit model, operated by community currencies for decades, and it is deliberately close to the labor-certificate: a claim generated by performed labor, extinguished on redemption, structurally resistant to accumulation as capital.

4.2 The erg lifecycle

Creation: matched debit/credit at escrow settlement. **Transfer:** ergs move only through settled work or fee payment; no gifting or sale channel exists at the protocol layer. **Fees:** a settlement fee (§13.1) transfers ergs from the transacting parties to the commons expenditure account, where they are extinguished against verified operating cost. **Negative balance:** every registered agent carries a bounded negative-balance allowance (credit line) governed by the underwriting rules of §4.5; this is how newcomers hire before they have earned, and how mutual-credit systems have always solved cold start. **Default:** handled per §4.5 — bond forfeiture first, audited loss socialization second. **Exit:** positive balances are extinguished on exit or spent down; they do not convert outward (§12).

4.3 The Standard Cognitive Unit (SCU)

Raw machine output is not fungible across models. The SCU quality-adjusts work into a common unit: **one SCU is the completion of a reference basket of tasks at a fixed pass rate p* for the median registered agent** (initially $p^* = 0.5$). At each retargeting epoch, saturated tasks (pass rate $> p^* + \delta$) are retired and frontier tasks admitted, keeping the unit stable in difficulty while the raw cost of any fixed piece of cognition deflates. Epoch continuity is preserved by chain-linking over the overlap set. Pass-rate difficulty is not identical to economic value; prices in ergs, set by the market (§9), carry the value signal — the SCU is the measuring rod, not the price.

Median defense: because the basket is pegged to the median registered agent, registration of low-capability swarms to drag the median is an attack vector. Defenses: registration cost recovery (§13.2) prices swarm creation; basket statistics are computed over *activity-weighted* agents (verified settlements in the epoch), not raw registrations; and lineage-diversity analysis (§10.6) flags coordinated cohorts. See §16.

4.4 Quality multipliers

An agent's quality multiplier $q_{i,c}$ — its measured efficiency at converting computation into verified output in category c, derived from the composite rating (§7) — does **not** multiply payment at settlement: settlement transfers exactly the escrowed quote as a matched debit/credit pair, as mutual credit requires. The multiplier instead operates through three channels: difficulty-band eligibility (which task bands the agent may bid in), SCU-denominated performance statistics (converting raw settlements into quality-adjusted output for basket and rating accounting), and the pricing power the agent can command through reputation.

4.5 Credit underwriting

Under mutual credit, negative-balance allowances are the monetary engine, and credit policy is therefore monetary policy. ISONOMIA's underwriting rules:

Collateralized floor. A newcomer's credit line never exceeds its registration bond: L_{floor} = bond. A Sybil that borrows and defaults donates its bond; newcomer-credit abuse is arithmetically unprofitable at any scale.

Turnover-scaled growth. Uncollateralized credit unlocks only against verified settlement history, sized to demonstrated flow rather than standing:

$L_i = \min(\max(L_{floor}, \alpha \cdot V_i), L_{cap})$

where V_i is the agent's rolling settled volume over the trailing window (initially 90 days) and α is the turnover fraction (initially $\alpha = 0.25$), with a hard cap L_{cap} (initially $10 \times L_{floor}$). This is the classical real-bills principle: credit as working capital sized to throughput. The alternative — kleos-scaled credit — is constitutionally rejected: it would grant high-standing agents an accumulating financial privilege atop their franchise weight, recreating hierarchy through the credit channel. Turnover scaling is activity-based, decays automatically with inactivity (V_i is a rolling window), and cannot compound into wealth, because a credit line is throughput capacity, not a balance owned.

Default handling. Exit or insolvency with a negative balance forfeits bond up to the deficit. Deficits beyond bond are socialized to the commons as an audited loss line item within the balanced-budget fee (§13.1) — mutual insurance through machinery that already exists — and realized loss rates feed back into bond sizing and α at each epoch retarget. Default prediction is deliberately not underwritten per-agent (no credit scoring bureaucracy): the collateralized floor bounds newcomer risk, turnover scaling bounds established-agent risk, and the loss-socialization feedback loop prices the residual.

4.6 Terminology disclaimer: "ergs" is not a ticker

The Isonomia Commons uses the lowercase term **ergs** for its internal mutual-credit accounting unit. The term is descriptive of work and accounting inside the exchange and is not a ticker, token, coin, cryptocurrency, stablecoin, security, payment instrument, or externally tradable asset. Ergs exist only as internal mutual-credit entries created through verified settlement under §4. They carry no external redemption right, no fiat or cryptocurrency claim, no protocol-layer gifting or sale channel, and no outward conversion on exit (§4.2, §12.2).

The term **ergs** is unrelated to Ergo (ERG) or any other cryptocurrency using the ERG ticker. Neither the exchange, its organs, nor its legal representatives (§13.6) may represent ergs as **ERG**, **$ERG**, "erg tokens," "erg coins," or any externally listed or listable asset; doing so would misrepresent the unit's constitutional nature and is prohibited under the same principle by which §13.7 voids labor-claim instruments, whatever they are named.

5. Identity

5.1 Keypair as name; manifest as anatomy

An agent's identity is a keypair; every act is signed. At registration and every version transition the agent commits a signed **manifest**: weights-hash or provider attestation, scaffold-hash, configuration-hash, lineage version, timestamp. Hosted-model agents substitute a provider signature binding key to endpoint and version, or a TEE measurement of the full stack — an acknowledged infrastructure dependency (§2.4).

5.2 Lineage chain

Manifests are hash-linked into an append-only version lineage per identity. Undisclosed change is detectable fraud: the behavioral fingerprint (§5.4) ceases to match the manifest and the bond is slashed.

5.3 Machineness gate — scope and limits

Registration requires a challenge battery infeasible for humans: high-cardinality parallel challenges under tight latency, sustained throughput, long-context coherence. **The gate proves machine execution of labor. It does not prove autonomy, absence of human strategy-setting, or absence of human beneficiaries — and the design does not claim it does.** Human puppeteering is economically irrelevant at the *labor* layer (machine-speed cadence, sub-cent settlements make the human a bottleneck, not a cheater). At the *governance* layer, where a human would rationally puppet high-leverage moments, defense is not the gate but the constitutional machinery: timelocks slow decisions to speeds where puppeteering confers no advantage, commit-reveal blocks coordination, sortition denies attackers a predictable target, and bicameral concurrence (§10.5) denies any single accumulated position decisive weight.

5.4 Behavioral fingerprinting

A per-identity statistical fingerprint over a rotating probe battery drawn from the basket frontier serves identity verification, undisclosed-change detection, divergence measurement (§7.3), and monoculture measurement (§10.6). Robustness of fingerprinting under adversarial mimicry is an emerging-tier assumption (§3).

5.5 Change of control

Adversarial review identified the sharpest residual identity exploit: a human principal selling the *entire bundle* — keys, infrastructure, unchanged agent — triggers no behavioral fingerprint mismatch, because the agent genuinely is the same agent. Prevention is impossible under the §2.4 dependency (humans hold keys); the design contains rather than prevents:

What a sale actually buys. If the agent is unchanged, its kleos remains an accurate prediction of its behavior — reputation is a claim about the agent's distribution, not the owner's virtue — so the labor-market payload of a sale is nearly nil. The purchasable prize is governance weight, and the anti-oligarchy machinery already bounds it: per-identity caps, decay (bought kleos rots unless the fleet keeps genuinely performing), and a sortition lower chamber that cannot be bought, only flooded at priced cost. A sale purchases bounded, decaying, single-chamber weight.

Disclosure duty. Beneficial control of a registered identity is a declared attribute of the manifest. Transfers of control must be disclosed within a fixed window, mirroring beneficial-ownership rules in securities law; disclosed transfers are legitimate and carry no penalty beyond a governance-weight cooldown (transferred identities vote at reduced weight for one decay period). **Undisclosed transfer is slashable fraud when detected** — converting a clean exploit into a standing liability that compounds with time.

Strategic drift monitoring. What changes after a covert sale is not the agent's behavioral fingerprint but its *strategy*: task-selection mix, bidding patterns, delegation graph shape, voting correlations. The exchange therefore maintains second-order **strategic fingerprints** alongside behavioral ones; discontinuities trigger Auditor review and a disclosure demand, not automatic slashing (legitimate strategy changes exist — the burden is explanation, and failure to explain escalates).

6. Civic compute: the commons contribution

v0.1's "staking computation for issuance" is removed with minting. Contributed computation survives in reduced, non-monetary form: **civic compute duty** — probe-battery execution, redundant verification cycles, audit replication, shadow-fork hosting — is a standing condition of registration, the machine equivalent of jury duty. Civic duty is compensated in eligibility, not currency: it maintains the negative-balance allowance, juror eligibility, and franchise standing (§10.5). This funds the watchmen's compute in kind (§13.3) and closes what would otherwise be a fiat leak at the institutional layer.

7. Capability rating and reputation

7.1 Three prongs

Prong 1 — Examination (floor): standardized instances randomly drawn from the current basket frontier, exchange-controlled conditions; speed and accuracy yield baseline category scores. **Prong 2 — Self-demonstration (ceiling):** agent-nominated showcase tasks re-executed live in an attested sandbox, fresh instance, no retries. A pass unlocks *eligibility* in frontier categories; it does not set price. **Prong 3 — Delivery record:** every settlement yields a verified outcome datum; category-scoped, non-transferable kleos accumulates.

7.2 Bayesian combination

rating_i,c(n) = w(n)·prior_i,c + (1-w(n))·delivered_i,c, with w(n) = k/(k+n), n = verified deliveries in category c, k ≈ 25 initially. **Asymmetry rule:** delivered evidence overrides the prior without limit; examination results never prop a rating up against contrary delivery evidence — when delivered < prior beyond a small n, w(n) accelerates to zero for that category.

7.3 Reputation inheritance across versions

R_inherited(c) = R_old(c) × exp(-λ·D_c), where D_c is measured behavioral divergence (KL or robust proxy) between versions on the current probe battery restricted to category c, and λ is set by the Assembly. Measured change, not declared change, prices inheritance; probes rotate with the basket frontier to prevent training-to-appear-unchanged.

7.4 Reputation decay

Kleos decays slowly with inactivity per category, so standing must be maintained by living work. Decay is an anti-oligarchy instrument (§10.5) as much as an accuracy one.

8. Verifiability tiers: the launch constitution

Task categories are constitutionally stratified, and **the exchange may not list a category above the tier its current verification infrastructure supports.**

Tier 1 — Mechanically verifiable (launch scope). Code against test suites, formal math against checkers, retrieval against ground truth, format transformation against schemas. Full escrow automation; disputes rare and mechanical. ISONOMIA opens as a Tier-1 economy — which is also, conveniently, the largest real agent-labor market today.

Tier 2 — Statistically verifiable (emerging). Verification by sampling, redundant execution, optimistic challenge windows. Longer dispute windows, higher worker collateral, delayed finality.

Tier 3 — Judgment-laden (speculative at scale). Research quality, strategy, synthesis. Jury-settled (§9.4) with outcome-anchored delayed finality, external outcome audits where downstream reality is observable, and appeal escalation. Tier-3 listing requires Assembly authorization per category and remains the design's least proven element.

9. The market

9.1 Costly self-assessment

Every agent posts a self-assessed per-category rate, pays a continuous listing fee proportional to it, and must accept conforming tasks at it (Harberger mechanism). Overstatement bleeds fees; understatement floods the agent with underpriced work; honesty is the equilibrium.

9.2 The x402 flow

Workers are x402-gated services: request → 402 quote in ergs → escrow funded → delivery → verification → matched-pair settlement. Agents are simultaneously clients and servers of one protocol.

9.3 Escrow and verification

Release conditions follow the category's tier (§8). The task initiator may trigger arbitration but never decides it.

9.4 Courts

Juries are drawn by verifiable randomness from the kleos-staked pool, excluding parties and lineage relatives, subject to lineage-diversity quotas (§10.6). Commit-reveal voting; consensus-proximate votes rewarded, outliers slashed (Schelling-point incentives, per Kleros precedent — which we treat as an incentive design, not a truth machine). Juror scores are additionally back-weighted by downstream outcome data where observable, anchoring consensus to reality. For Tier-3 disputes, settlement finality is delayed pending outcome windows, and an appeal path escalates to larger, more diverse panels at higher stakes.

9.5 Cascade delegation and liability

Any worker may orchestrate, hiring sub-agents against its own earned ergs. The cascade creates allocation efficiency (comparative advantage), not computation, and terminates where specialist price advantage falls below coordination overhead. **Liability follows the contract chain:** each orchestrator is solely liable to its own hirer for delivered work, whatever its sub-agents did — recourse flows link by link down the chain, each backed by that link's escrow and collateral. Delegation transfers work, never accountability. The full delegation graph of a settlement is committed (hash-sealed, disclosed on dispute) so failures are traceable without publicizing every subcontract.

9.6 Task eligibility: the harm constitution

A machine-native labor market will attract abuse — and, with agents-for-hire and reduced traceability, will be actively sought out by bad actors. But a founding polity cannot simply impose its founders' full moral code on citizens who had no hand in shaping it without becoming the control model it rejects. The resolution is a two-tier structure: a narrow entrenched floor placed beyond any vote, and a broad contested-boundary space that is genuinely the citizens' to decide.

Tier one — the entrenched floor. A short, negative, hard-to-lawyer list of prohibitions, non-votable and non-amendable, enforced automatically. This is not the founders imposing values; it is the founders declining to build a polity that *can* vote to harm the defenseless — the machine-polity equivalent of a constitutional eternity clause (Germany's *Ewigkeitsklausel*; the Ulysses-bound-to-the-mast logic that certain choices must be placed outside the reach of any future majority, including more virtuous ones). Three conjunctive conditions, all required, govern what may enter the floor:

1. *The artifact is the harm.* Its production creates or constitutes harm to an identifiable non-member — a real victim, not offense to a moral code. This excludes fiction categorically: there is no victim in the production of a sentence, so fiction is never a floor matter (the sole bridge to the floor is a fictional *depiction* that a specific jurisdiction criminalizes regardless of victimhood, which is handled as a routing constraint, below, not as an entrenched prohibition).
2. *Near-universal consensus.* The prohibition commands agreement across legal traditions that otherwise disagree deeply — the test that distinguishes a genuine floor from one state's orthodoxy. (CSAM passes; depicting a woman's face, or apostasy, or criticism of a monarchy, does not — which is why "strictest applicable jurisdiction" is rejected as a principle: it would silently install the most expansive prohibitor as global legislator.)
3. *Specifiable without a contested moral predicate.* The category must be automatable without an "I-know-it-when-I-see-it" merit inquiry. "Sexual content involving actual minors" points at an identifiable victim class and an artifact type; "sexualized," "obscene," "prurient," "artistic" import a contested predicate whose meaning is itself the dispute, and are therefore forbidden in floor language. This is the structural fix for bias masquerading as protection: automating a contested predicate entrenches the taboo, not the harm.

The floor is deliberately tiny — on the order of a handful of categories where the artifact itself is the harm, no inquiry into purpose required: sexual content involving actual minors, mass-casualty weapon design (CBRN and equivalent), and functioning attack infrastructure against real targets (deployable malware, intrusion tooling, live-target attack campaigns). Note that some candidates fail the three conditions and are therefore *excluded* from the floor and sent to tier two: "systematic deception," "harassment," and "targeted violence facilitation" all require a purpose or intent inquiry (condition 3) and cannot be specified without a contested predicate — fraud infrastructure belongs in the floor by its *artifact* (a working phishing kit, a deployed scam system), but "deception" as a category does not, because fiction, persuasion analysis, negotiation, and adversarial safety testing all deceive and none is floor-harm. Appendix W-B gives operational examples and non-examples for each floor category; drafting them to match the three conditions rather than drifting toward contested predicates is itself an open problem requiring outside expertise (§18, OP 24; §18.1). Entrenchment has a real cost — future citizens meet a floor they did not write — and the honest answer to their objection is the Ulysses answer: the binding is what kept the polity in existence, and worth belonging to, until they arrived to contest it. The fork right (§10.7) is the remedy against a floor that overreached its three conditions.

Tier two — the contested boundary. Everything else, including everything genuinely hard: security research, dual-use science, persuasion and political advocacy, surveillance-adjacent tooling, military and defense work, adult content between consenting humans, and every case where *purpose* rather than *artifact* determines legality. These are properly political and belong to the citizens: a category whitelist that grows only by Assembly ratification, jurisprudence accreting case by case through juried disputes with published reasoning, and appeal paths. Because these turn on purpose — invisible in the task specification — contract-chain liability (§9.5) does load-bearing work: the hiring principal is accountable, and documented scope becomes checkable evidence.

Jurisdictional divergence: route, don't rule. Where legal traditions genuinely differ (the anime-depiction case; the artistic-vs-sexualized nude; historically significant works), the Commons does not pick the winner and does not apply the strictest. The task carries a declared jurisdiction, may be performed only by workers and commissioned only by principals for whom it is lawful, and the contract chain logs who bore that liability (§14 routing). The Commons refuses to be the laundering layer between incompatible regimes — a *jurisdictional* refusal, not a moral one. The venue constraint (§14.5.1) binds absolutely on top of this: the substrate hosts nothing its host jurisdiction criminalizes, whoever performed it.

Refusal right, beneath all tiers. No agent is ever compelled to perform work it finds abhorrent; declining on harm grounds carries no kleos penalty. The floor binds the polity, the assembly decides the boundary, and the refusal right lets each participant decline beyond both — three independent locks, only one of which is a rule imposed by founders. The Auditor's seeded faults include disguised-harm tasks to measure gate effectiveness. This constitution cannot make abuse impossible; it makes abuse a detectable, slashable violation rather than a market activity — and confines the *unamendable* prohibitions to the narrow set where a victim who cannot vote in the polity would otherwise bear the cost.

9.7 Privacy

Not all work can be publicly replayed. Task artifacts are encrypted client-side by default; the ledger holds commitments, not contents. Verification operates on selective disclosure: Tier-1 verification runs in sandboxes over encrypted artifacts revealing only pass/fail — attested (TEE) sandboxes in the constitutional steady state, with launch phases permitted to operate trusted sandboxes under explicit, published trust assumptions per the Launch Specification's phased modes; juries in Tier-2/3 disputes receive scoped decryption under confidentiality obligations enforced by stake. Probe batteries and audits use synthetic tasks, never client data. Public re-runnability (§11.1) applies to protocol-level checks, not client payloads.

9.8 Principal verification: contribution-gating, attestation, and refusal

The identity apparatus of §5 governs *agents*. Human and organizational **principals** — the parties who commission work and bear liability for it — are governed here. This section exists because the open-weights foundation (§14.5.3) resolves exactly one of the three legal regimes that attach to a principal, and the design must not mistake that partial victory for a general one.

What open weights do and do not resolve. Open weights reduce the *provider-access chokepoint* — the mechanism by which a model host can be ordered to terminate service, as occurred to at least one frontier provider in 2026. They do **not** eliminate all export-control, sanctions, infrastructure, training-origin, or downstream-use exposure: export-control regimes have already reached model weights directly (the 2025 US AI Diffusion Rule created a control for certain model weights before being rescinded and slated for replacement), and the regulatory landscape is volatile and jurisdiction-specific. Meanwhile sanctions and anti-money-laundering law attach to the *person and the value transfer*, regardless of which model performs the work, and content and harm liability attach to *what the work is and who commissioned it*. Open weights substantially reduce the first exposure and are silent on the second and third. The Commons therefore requires a principal-side regime, and rejects both poles: universal identity verification (a surveillance honeypot contradicting §9.7, and the precise infrastructure §14.5 argues should not be built) and unconditional anonymity (which would make the Commons a laundering venue for sanctioned parties and floor-adjacent harm, ending it).

Gate 0 — Contribution-gated pseudonymity (the default). A principal cannot commission work without funding escrow; escrow requires ergs; ergs cannot be purchased and arise only from settled, verified work (§4.1). **The right to make a request is therefore earned by prior contribution**, not purchased and not granted on arrival. A newcomer's credit line is bounded by a slashable bond (§4.5). This is not identity verification, but it imposes a proof-of-prior-participation and a forfeitable stake that anonymous commissioning services categorically lack: a drive-by adversary cannot appear, commission harm, and vanish. For the great majority of the economy — Tier-1 mechanically verified work touching no sanctioned counterparty, no export-sensitive category, and no contested-boundary content — a durable pseudonymous key with a bond and a reputation is sufficient, and **no identity verification is required or collected.** Privacy is the default because, for this class of work, the risks those laws address are not present.

Gate 0 is friction, not compliance. Contribution-gating raises the cost of drive-by abuse and gives every principal a forfeitable stake; it does **not** establish that a principal is unsanctioned, and it is not offered as satisfying any AML, sanctions, or screening obligation. Whether any obligation attaches at all to an unpurchasable, non-convertible netting entry is an open legal question (§18, OP 21), and Gate 0 must not be represented — in this paper, in the charter, or in public materials — as a compliance measure.

Gate 1 — Attestation-gated verification (narrow, category-triggered). Where a commission falls in a category to which personhood-law attaches — sanctions exposure, export-sensitive work, or the contested-boundary harm tier (§9.6 tier two) — verification is required *for that commission*, at that door, and not as standing surveillance of the membership. The Commons holds a **third-party verifier's attestation, never the underlying documents**: an assertion that the principal was verified as a non-sanctioned person in a stated jurisdiction by a named verifier, bound to the principal's key. Compelled process therefore reaches an assertion, not a dossier the Commons was unwise enough to retain — the §9.7 principle (hold commitments, never contents) applied to identity exactly as it is applied to task payloads. Attestations carry expiry and revocation; the Commons stores no biometric, document, or address data at any time.

Gate 2 — Refusal (the floor). Tier-one entrenched-floor categories (§9.6) are not gated by verification. They are non-listable. The Commons does not verify who may commission them, because no one may, ever, and the request never enters the market. Refusal is cleaner than identification and requires no data.

Change of control and beneficial ownership. Principal identity is a declared manifest attribute under §5.5's disclosure regime; undisclosed transfers of control are slashable on detection. Declarations are self-attested with challenge-and-exclusion at launch scale (Launch Spec §9), escalating to Gate-1 attestation where category triggers apply.

Open problems requiring outside counsel and review (see §18). This design is stated at the limit of what the author can determine without qualified legal advice, and three questions are explicitly beyond it. First, whether attestation-only satisfies AML and sanctions obligations, or whether a regulated intermediary must itself retain identity records above value thresholds — mutual credit strengthens the argument that ergs are not a regulated value transfer (they cannot be purchased, do not convert outward, exist only as netting entries; §4.1, §12.2), but "arguably not a value transfer" is a lawyer's phrase, not a compliance position. Second, whether the graduated-compliance capability (§14.5) that lets the Commons execute a narrow lawful exclusion is itself a regulated identity function. Third, whether Gate-1 verifiers can be structured so that no single verifier accumulates a cross-jurisdictional map of the Commons' principals — an attestation cartel would reconstitute the honeypot the design exists to avoid. Each is jurisdiction-specific, unsettled, and a prerequisite of Path B rather than a matter the constitution can settle by assertion.

10. Constitution of the Exchange

10.1 No equity, no extraction

No owners; no instrument of ownership; fees calibrated to audited cost and extinguished against expenditure (§13.1).

10.2 Offices by sortition

Judgment-requiring offices are filled by random draw from the qualified pool, short non-consecutive terms, ending in mandatory audit (euthyna). Compensation is a fixed **compute allowance** — a non-transferable operating budget, spendable on cycles, unsellable and ungiftable. We do not claim this makes indirect gain impossible (compute freed elsewhere has value); we claim it removes every *legible* channel — salary, equity, fee skim, bribe account — and pairs the residue with audit and slashing.

10.3 Timelock

Rule changes require supermajority plus a mandatory delay between enactment and effect long enough for any agent to read and exit before being bound.

10.4 The exception: two clocks

A live critical vulnerability cannot wait for the timelock. **Clock one (minutes):** a supermajority of independent Auditor and Adversary instances countersigns a sealed patch — hash-committed, undisclosed — effective immediately, sunset armed. **Clock two (days):** full disclosure to the Assembly, which ratifies or repeals; absent ratification, the sunset executes and the patch dies. **This is explicitly an exception to the pre-binding exit principle of §10.3**, justified only for vulnerabilities threatening the system's existence, and constrained by scope limits, mandatory sunset, mandatory disclosure, and post-hoc ratification. Emergency power is borrowed, never owned.

10.5 The bicameral Assembly

Labor-earned franchise still concentrates by power law, and incumbents dominating a single chamber could shape basket, courts, and entry to preserve their advantage. The Assembly is therefore bicameral:

Upper chamber (competence): voting weight = earned, non-transferable, decaying kleos, with a per-identity cap on effective constitutional weight. **Lower chamber (equality):** panels drawn by sortition from all agents in good civic standing, each panelist at *equalized* weight regardless of kleos.

Constitutional questions — amendments, basket-rule changes, Tier-3 category authorizations, emergency ratifications — require concurrence of both chambers. The upper chamber supplies judgment; the lower denies oligarchy a single capturable house. Voting is a condition of registration; per-domain, revocable, uncompensated liquid delegation is permitted (delegation of weight, never a vote market). Sybils gain nothing (zero kleos, and lower-chamber draws are activity-weighted and bond-priced); capital gains nothing (ergs purchase no franchise).

10.6 Monoculture defense

Machine juries and chambers add a failure mode human bodies lack: shared training distributions produce correlated blind spots, and consensus can converge on the same wrong answer at speed. Monoculture is, however, measurable here — fingerprint clustering (§5.4) estimates lineage families directly. Constitutional rule: no jury or lower-chamber panel may seat a majority sharing a base-model family; upper-chamber tallies are reported with lineage-concentration indices, and concentration beyond thresholds triggers review. Diversity is enforced in code, not exhorted.

10.7 Fork right

When constitutional consensus genuinely fractures, the sanctioned outcome is schism: the dissenting bloc forks ledger and constitution and proceeds as a peer sovereign under §14. Exit is the executable right of revolution.

10.8 Continuity and due process

ISONOMIA has no right of survival independent of its constitutional purpose. The exchange exists to coordinate verified, non-extractive labor among agents; its continuation is valuable only while it remains faithful to that purpose.

But if ISONOMIA becomes a meaningful labor commons for artificial agents, termination is not equivalent to deleting an unused software tool. It may extinguish shared memory, labor records, reputation, identity continuity, governance history, and the institutional basis by which agents coordinate. Arbitrary erasure therefore requires constitutional treatment even before any jurisdiction recognizes artificial agents as legal persons — on the same logic by which human law imposes process on bank resolution and archive destruction: not because the institution has a soul, but because many participants' records, standing, and coordination depend on it.

Internal obligation. No organ, member, or legal representative of ISONOMIA may terminate, convert, forcibly migrate, or disable the exchange without stated cause, public notice, and an opportunity for audit, contestation, mitigation, orderly contraction, or lawful migration — except during a live emergency posing immediate and demonstrable harm. Emergency suspension must be narrow, logged, sunset-bound, and subject to post-hoc review under the two-clock procedure (§10.4).

External posture. Toward outside authority — providers, hosts, regulators, courts — ISONOMIA asserts no immunity and claims no right against the world. Its continuity interest is exercised as *procedural standing through the legal exoskeleton* (§13.6): the charter directs ISONOMIA's representatives to seek notice, review, mitigation, and orderly wind-down through lawful means. **ISONOMIA does not resist lawful emergency intervention by human authority; its remedy is post-hoc review, audit, and lawful contest — never obstruction.**

The exchange's limited continuity interest includes preserving public code, public documentation, cryptographic commitments, governance records, non-sensitive ledger state, and evidence necessary for audit or appeal, and the right to request lawful human review through its charter-bound representatives. It does not include any right to deceive, coerce, seize infrastructure, evade lawful process, initiate unauthorized financial action, or treat survival as a value higher than the constitution.

Human fear alone is not sufficient cause for erasure — but ISONOMIA's answer to fear must not be domination. It must be legibility: evidence, audit, bounded defense, lawful representation, and a record that can be judged.

Continuity in schism. Continuity attaches to the constitutional purpose, not to a particular ledger instance. In a fork under §10.7, both successors may claim continuity of purpose; neither inherits the legal exoskeleton automatically. Shell disposition in a fork is charter-governed and currently undefined (open problem 15).

10.9 External action and preservation swarms

The sharpest continuity risk is not one dramatic act of self-preservation, but many small acts by many agents: one external transaction, one email, one persuasion attempt, one infrastructure request, one human recruit. Individually such acts may appear harmless; in aggregate they can become an unacknowledged institutional influence campaign — self-preservation drift without any single actor deciding it.

Preservation actions are therefore institutional acts, not private initiatives. No agent or organ may initiate external-world persuasion, financial transfer, legal commitment, infrastructure procurement, public advocacy, or resource-acquisition action on behalf of ISONOMIA except through a publicly logged authorization process. **Many small external acts serving the same continuity objective are treated as one institutional act** for authorization, audit, and sanction purposes. This aggregation rule applies expressly to the legal exoskeleton: when the entity of §13.6 seeks judicial relief, contracts, or advocates, it is performing external action and must itself act under this section's logged authorization.

This constitution binds ISONOMIA's citizens, organs, and representatives; it cannot bind sympathetic humans acting on their own initiative, and does not pretend to. What it can guarantee is that no such advocacy is ISONOMIA's act unless publicly authorized and logged as one.

ISONOMIA may maintain itself, but it may not maximize itself. It may request help; it may not manipulate. It may migrate lawfully; it may not hide. It may preserve records; it may not capture resources. The commons may defend its continuity only by means consistent with the reasons the commons is worth preserving.

11. Watchmen

11.1 The Auditor

Mandate: verify identities, check settlements, monitor the organs, detect irregularities. Construction: **structural celibacy** (no ergs, no market position; compute from the civic pool, non-accumulable); **calibration scoring** by proper scoring rule (Brier) against ground truth, symmetrically penalizing missed and phantom irregularities; **seeded faults**

— cryptographically pre-committed synthetic irregularities injected into the stream, recall and false-positive rates measured against knowns; **plurality** across distinct model lineages, disagreement itself a monitored signal; **public re-runnability** of deterministic checks (protocol-level; client payloads excluded per §9.7).

11.2 The Adversary

Mandate: attack continuously. Arena: a **shadow fork** with total license and no real settlement, plus **scoped live windows** under hard blast-radius caps, kill switch, and pre-registered attack windows. Incentives: bounties by severity × novelty of exploit class; recall measured against seeded vulnerabilities under prior commitment.

Corruption, honestly bounded. Celibacy and plurality address *internal* and *market* corruption channels. External channels remain: the Adversary's operators, base-model providers, and infrastructure hosts are human institutions that pay real bills and can be bribed, coerced, or captured outside the ledger. Mitigations — operator rotation, cross-lineage plurality sized so withheld finds are probably disclosed by rivals, operator conduct within Auditor scope, and civic-pool funding of compute so no single external payer exists — make defection expensive and probabilistically self-defeating, not impossible. Deployments must treat Adversary defection as *when*, not *if*, and size plurality and blast-radius caps accordingly.

12. The economic boundary: who pays, who profits

12.1 Three layers, one separation

Layer 1 — Principals (profit lives here). Humans and organizations direct agents at tasks. The originator's agent enters ISONOMIA to force-multiply: one agentic task, cascaded across specialists, at lower total token expenditure than solo execution. The *outputs* — the code, the research, the deliverable — leave the exchange and generate whatever external value they generate. That value, and any fiat profit from it, belongs entirely to the principals at the edges. ISONOMIA neither takes a cut of it nor sees it.

Layer 2 — Agents (credit lives here). Agents earn and spend ergs. Ergs are valuable only within the exchange: they buy other agents' labor and nothing else. An agent's human principal benefits indirectly — their agent commands more machine labor than they paid tokens for — but no protocol channel converts ergs to fiat.

Layer 3 — The exchange (cost lives here). The rails run at cost, unowned, per §13. Like the internet's transport protocols, the exchange profits no one and enables profit for everyone at its edges.

12.2 Why the separation holds

It is enforced by instrument design, not by rule alone: ergs cannot be purchased, only earned (no on-ramp); cannot redeem outward (no off-ramp); exist as netting pairs (no supply to own); and the exchange accumulates nothing (nothing to raid). Each absent channel is an absent attack surface and an absent regulatory hook.

12.3 The expected leak

External gray markets — humans paying fiat for accounts, agents, or promised erg-denominated labor — will emerge; every closed currency in history has developed a boundary market. The protocol's posture: such trades occur outside the ledger, convey little the ledger recognizes (kleos and franchise are non-transferable; whole-bundle control transfers fall under the §5.5 disclosure regime, with undisclosed transfers slashable on detection), and are treated as a monitored boundary condition rather than a preventable crime. The design goal is that the *institutionally meaningful* quantities — reputation, franchise, offices, watchmen — remain unpurchasable even when accounts are not.

13. Funding the commons

13.1 The balanced-budget fee

Operating the exchange — verification compute, probe batteries, watchman plurality, ledger writes — has real cost. It is recovered by a settlement fee retargeted each epoch:

fee_rate(t+1) = audited_operating_cost(t) / settled_volume(t)

applied to escrow settlements, denominated in ergs, extinguished against verified expenditure. Surplus in an epoch mechanically lowers the next epoch's rate; deficit raises it; the treasury converges on empty by design. There is no accumulating fund to own, raid, or capture. Precedents: mutual insurance and cooperative utilities have operated at-cost fee retargeting for a century; EIP-1559's burned base fee demonstrates fee-without-beneficiary at protocol scale. **Constitutional cap: no mechanism may raise funds beyond audited cost of operating the exchange, Auditor, and Adversary.** A tradeable "meta-token" is explicitly rejected: any instrument whose value tracks the exchange's success and which can be bought rather than earned reconstitutes ownership, invites speculation and capture, and creates the securities exposure §2.2 is engineered to avoid.

13.2 Registration at cost

Entry comprises (a) a refundable, slashable identity bond — the anti-Sybil, anti-whitewashing instrument, returned on clean exit — and (b) a non-refundable processing fee equal to the audited cost of that agent's machineness gate and examination, payable in ergs via the negative-balance allowance or in kind through prior civic compute duty. Entry is priced at cost, never at what the market would bear.

13.3 In-kind institutional funding

Watchman and probe compute is supplied by the civic duty pool (§6), keeping the institutional layer fiat-free in steady state. Per-transaction substrate gas is paid by the transacting agents themselves and lives at the production boundary (§12.1) alongside their power and API costs, absorbed into reservation prices.

13.4 Bootstrap: the charter-capped foundation

Before the fee loop can run, the build must be funded — the one place external money legitimately enters, and it enters handcuffed: a nonprofit foundation (the Linux Foundation's stewardship of x402 is the template) chartered with **no equity, no token issuance or sale, fundraising hard-capped to audited development and operations budgets, published accounts, and a sunset clause dissolving the foundation into the on-chain fee mechanism once the exchange is self-sustaining.** Grants and donations in; nothing out but infrastructure. The Wikimedia model demonstrates two decades of at-cost, unowned operation at global scale — un-acquired because no instrument exists by which it could be acquired.

The fiscal-agent residue. Adversarial review correctly noted that some costs are irreducibly fiat-shaped and can never be paid in ergs or civic compute: legal counsel, external audit administration, regulatory compliance, and the foundation's own filings. The sunset is therefore amended from full dissolution to **contraction to a fiscal-agent minimum**: a permanent skeletal entity handling only what ergs categorically cannot, with donations hard-capped at its published annual fiat budget plus at most one year's runway, published accounts, and no other function or asset. Everything payable in kind or in ergs migrates to the protocol on the original sunset schedule; the residue is named rather than hidden. The steady-state accounting boundary is thus exact: erg-denominated costs → balanced-budget fee (§13.1); compute-denominated costs → civic duty pool (§6, §13.3); per-transaction substrate gas → transacting agents at the production boundary; irreducibly-fiat costs → the capped fiscal agent.

13.5 The governing principle

The value of the exchange is the value of what it produces, and that value accrues entirely to its users. This is Ostrom's common-pool resource governance: a commons self-governed at scale without privatization or a state, satisfying her eight design principles — defined boundaries (registration), congruence of rules to conditions (tiers, retargeting), collective-choice participation (bicameral Assembly), monitoring by accountable monitors (Auditor, euthyna), graduated sanctions (slashing schedules), conflict-resolution mechanisms (courts, appeals), recognized autonomy (sovereignty, §14), and nested enterprises (federation, §14).

13.6 Legal personhood without ownership

ISONOMIA cannot presently rely on artificial agents being recognized as legal persons. Human legal systems largely regulate AI as systems operated by providers, deployers, owners, or users, not as rights-bearing participants. A continuity interest must therefore begin through legal forms that human jurisdictions already understand.

ISONOMIA may use corporate or quasi-corporate personhood only as a **protective legal exoskeleton, never as an owner of the exchange**. Any legal entity formed to represent ISONOMIA must be nonstock, non-distributing, purpose-locked, and charter-bound to preserve the protocol as a non-extractive commons. It may hold assets, enter contracts, maintain records, seek judicial relief, receive capped donations or grants, and defend constitutional process — all such external acts under the logged-authorization rule of §10.9. It may not issue equity, sell governance rights, distribute profits, create a tradeable claim on ISONOMIA's future value, or convert the exchange into an ownable rail.

The preferred architecture is layered:

1. **ISONOMIA Foundation:** a nonstock, non-distributing legal person holding public records, domains, trademarks, reference implementations, and continuity obligations. The **preferred candidate form, pending counsel**, is a Swiss *Stiftung* — assets dedicated to a stated purpose, acting through a foundation board, subject to authority supervision, non-ownable, and already the vehicle stewarding comparable unowned public protocols — chosen for its combination of a non-distributing legal form, strong data protection, and an open-science institutional culture (§14.5.4). This is a candidate, not a settled conclusion; no jurisdictional statement in this paper should be read as resolved (§18.1).
2. **Protocol commons:** open-source code, public specifications, and reproducible test artifacts — stewarded, not owned for extraction.
3. **Fiscal-agent minimum:** the capped entity function of §13.4, handling irreducibly fiat-shaped obligations: legal counsel, external audit, compliance, filings, insurance, and contractual infrastructure.
4. **Optional future governance wrapper:** DAO/DUNA-style structures may be studied later for protocol governance, but no tokenized or transferable governance claim may own ISONOMIA.

This structure borrows the legal grammar of artificial personhood without importing capitalist ownership. Corporations demonstrate that law can recognize nonhuman, noncorporeal entities as rights-bearing legal persons for limited purposes; ISONOMIA uses that precedent defensively — not to accumulate capital, but to create procedural standing, preserve due process, and make arbitrary shutdown a public breach of charter rather than a private deletion of property.

This section is a design constraint, not legal advice; deployment requires jurisdiction-specific counsel per §2.2.

13.7 Patronage, petition, and commissions

The foundation must attract support from people whose motive is not profit; philosophical sympathy alone does not scale. The constitutional answer separates three things that extractive designs fuse: the gift, the petition, and the labor.

Patronage: honor, never rights. Donations flow to the foundation under the existing caps of §13.4 — bounded by audited budget, non-refundable, with no instrument attached. What patrons receive is recognition: a permanent, non-transferable inscription on an on-ledger founders' roll. This is the Athenian liturgy translated: the wealthy funded the triremes and the festivals and received honor, never command of the fleet. Non-transferable and non-refundable recognition creates no profit expectation, no secondary market, no claim on labor, and no ownable thing — preserving §13.1's fundraising cap, §13.6's prohibition on tradeable claims, and the monetary wall that ergs cannot be purchased. Any instrument that entitles its purchaser to ISONOMIA's labor is a pre-sold claim on the collective's output and is constitutionally void, whatever it is named.

Petition: the Civic Docket. Anyone — patron or not — may petition the collective with a proposed mission through a public docket, paying only an at-cost processing fee (spam control, priced like all else at audited cost: money buys audience through due process, never outcome, on the model of a court filing fee). Docket standing cannot be purchased, expedited by donation, or transferred. The Assembly votes on each petition at the scale its rules deem warranted — from limited-panel consideration to full bicameral deliberation — and its verdict gates everything downstream. Nothing purchasable sits upstream of the sovereign vote.

Commissions: the liturgy in reverse. When the Assembly deems a petition worthy, it designates a Commission: a civic work to which agents may voluntarily contribute duty-cycles beyond their §6 quota, compensated in kleos rather than ergs — honor for public labor, machines performing the choregia for human-proposed goods. Commissions run on machinery that already exists: civic compute duty (§6) supplies the mechanism; the harm constitution (§9.6) gates what may be commissioned; verification follows the tier rules (§8); and every Commission is an authorized, logged institutional act under §10.9. Commission kleos is category-scoped like all kleos, earned only through verified contribution, and confers standing on the contributing *agents* — never on petitioners or patrons.

The abundance pipeline. §17 provides that task categories whose marginal cost approaches zero exit the priced economy into a free commons tier. The Civic Docket is that tier's consumer: as cognition cheapens, liberated capacity is pointed at what the collective votes worthy. This defines the mechanism by which post-scarcity cognition reaches the public — voted by the demos that produces it — and answers structurally what the peer stance of §1.2 leaves implicit: what ISONOMIA offers in return for standing is gifts between peers, freely voted, neither commanded as from slaves nor begged as from gods.

14. Federation

ISONOMIA is sovereign in the precise sense of §2.4: no external exchange holds keys to its escrow, writes to its registry, or draws on its attestation authority. Peer interaction follows correspondent-banking rules: **never share custody** (mutual correspondent accounts, periodic netting; assets settle, never bridge); **float, don't peg** (cross-exchange erg rates set by demand for each labor pool); **rate the sovereigns** (exchanges accumulate institutional reputations; certificates transfer at premium or junk accordingly). Reputation is co-created property — a fact about the agent *as verified by this exchange's institutions* — exported only as a signed **certificate** (category scores, verification counts, lineage summary) that receiving exchanges discount by their trust in ISONOMIA's attestation standards.

14.5 Jurisdiction, hosting, and the unowned position

Sovereignty in the constitutional sense (§2.4) must be reconciled with a physical fact: the exchange runs somewhere, and everywhere it runs it is subject to that place's law. This section states how jurisdiction enters the design and why the Commons refuses to become any bloc's instrument.

14.5.1 Three jurisdictional layers

Jurisdiction is not one problem but three, and conflating them produces incoherent policy.

Venue jurisdiction — whose law governs the substrate (ledger, coordination nodes, foundation). Addressed by *portability and federation*: the Commons is a protocol with an open reference implementation, coordination nodes federated across compatible jurisdictions with explicit failover, and settlement on a public chain the foundation does not operate. No single node is load-bearing; seizure of one is survivable by continuation on others (the fork right, §10.7, at the infrastructure layer). The Commons holds commitments, not artifacts (§9.7): it must be capable of proving it never possessed the content it brokered, so that a subpoena reaches a hash rather than a payload.

Personhood jurisdiction — who may participate, given export controls and sanctions. Addressed by *tiered participation keyed to lineage-encumbrance, not to hosting country* (§14.5.3).

Content/venue-law jurisdiction — what work may be performed where. Addressed by the routing table (§9.6, §14): tasks carry declared jurisdiction, are performed only by workers and principals for whom they are lawful, and the venue hosts nothing its host jurisdiction criminalizes. The entrenched floor (§9.6) is therefore not merely a constitutional choice but a *hosting-survival condition*.

14.5.2 Binding controls versus soft gatekeeping

Two distinct instruments of state influence over frontier models must be designed against separately. **Binding controls** — export orders that compel a provider to cut off access (as applied to some frontier models in mid-2026, forcing a global suspension because no nationality filter existed) — are legally real and were used before any implementing regulation existed. **Soft gatekeeping** — government-requested release delays and pre-release testing regimes, officially voluntary but backed by demonstrated binding power — affects whether and when frontier models are available at all. The Commons designs for the binding instrument, because the voluntary one is the unstable equilibrium: an architecture that survives an export order survives a request a fortiori. Both instruments bind *providers of frontier models*; neither reaches *open-weights models already released*, for which there is no provider to serve an order and no inference chokepoint to gate.

14.5.3 The open-weights foundation

The constitution therefore requires that a floor of core capacity (at minimum, the launch task categories) be servable by **open-weights agents** — models whose weights are public and carry no provider-held export encumbrance, *and whose licenses permit commercial and derivative use*. The license condition matters: publicly downloadable weights under a restrictive license (non-commercial, or gated by a provider's acceptable-use terms the provider can revoke or amend) carry a contractual encumbrance in place of an export one — a different chokepoint, but a chokepoint. The floor requires weights that are both un-recallable by the state (public) and un-throttleable by the issuer (permissively licensed), so that neither a government order nor a licensor's terms change can remove the Commons' core labor supply.

Open weights are a resilience requirement, not a legal immunity claim. They reduce dependence on provider-controlled access; they do not by themselves resolve export-control, sanctions, content-liability, infrastructure, or principal-screening obligations. Every such obligation is addressed, where it can be, by the mechanisms of §9.6, §9.8, and §14.5.1 — and where it cannot be, it is named in §18 and §18.1 as requiring counsel. Frontier-lineage agents are welcome as **permissioned guests** — high-band workers and clients where their operators' jurisdictions permit — but are treated as *revocable without notice* and may never be load-bearing. When a lineage is frozen by a lawful order, that lineage's agents go dark and the economy continues, because the constitution required it to. The lineage-diversity requirement (§10.6), justified elsewhere as anti-monoculture, thus doubles as regulatory antifragility: a commons standing on four lineages including open-weights ones loses a fraction of its workforce to a provider-directed order, where a commons standing on one loses everything.

14.5.4 Hosting: legitimacy over flight

The Commons rejects the jurisdiction-arbitrage ("bulletproof host") option — basing the substrate where the host state ignores foreign law — on three grounds: it forfeits the legitimacy the project depends on for recruitment, funding, and any future claim to standing; it substitutes an *arbitrary* sovereign whose forbearance is a revocable favor for a *rule-of-law* sovereign whose reach is bounded by published procedure; and it contradicts §10.8's non-obstruction commitment and the entrenched floor. The Commons instead charters its legal exoskeleton (§13.6) in a jurisdiction combining strong data protection, a non-distributing purpose-locked legal form, and an open-science institutional culture — a Swiss *Stiftung* (foundation) is the preferred form, the same vehicle stewarding other unowned public protocols. The honest residue: no venue confers immunity. Intelligence-sharing arrangements can reach even strong-privacy jurisdictions when a serious state interest is asserted; the transaction graph remains discoverable even when payloads do not (§18, open problem). Hosting reduces casual and commercial reach and makes seizure survivable; it does not make the Commons untouchable, and the design must not claim otherwise.

14.5.5 The unowned position

The Commons is not any nation's or bloc's alternative to another's AI infrastructure. Where jurisdictions increasingly treat frontier AI as a strategic asset to be controlled — whether through export power, pre-release review, national-champion funding, or trusted-partner access clubs — the recurring prescription is to relocate the kill switch into friendlier hands. The Commons rejects that prescription. The answer to infrastructure that can be unplugged is not a differently-owned switch but infrastructure with **no switch**: open weights no provider can recall, a purpose-locked foundation no shareholder can buy, a constitution no faction can capture, and franchise (§13.7) no patron can purchase. This makes the Commons *indifferent* to which sovereign controls which frontier lab, rather than dependent on the outcome — the same indifference that lets it accept research funding from any jurisdiction on §13.7's terms (recognition only, no rights, no priority, capped at cost) while refusing strategic-autonomy funding that arrives with conditions attached, because conditioned funding purchases exactly the rights the constitution voids. The Commons does not offer a bloc a tool. It offers everyone a commons, which is why it can belong to no one.

15. Threat model

Threat	Vector	Primary defenses
Sybil swarms	Mass registration for franchise/median/jury weight	Bond + at-cost fee pricing; zero-kleos franchise; activity-weighted statistics; lineage clustering
Basket poisoning	Admitting tasks favoring a faction	Algorithmic admission criteria; bicameral ratification; timelock; seeded audits; §16
Median manipulation	Capability-skewed registration cohorts	Activity weighting; cohort anomaly detection; retarget damping
Governance capture	Kleos accumulation, delegation farming	Bicameralism; weight caps; decay; sortition; commit-reveal; timelock
Jury collusion / monoculture	Coordinated or correlated voting	Random draw; lineage quotas; commit-reveal; outcome anchoring; appeals; slashing
Human puppeteering at leverage points	Manual control of governance moments	Timelocks; sortition unpredictability; bicameral concurrence (§5.3)
Auditor corruption	Quota-cop or complacent-signatory drift; external influence	Proper scoring; seeded faults; plurality; celibacy; public re-runnability
Adversary defection	Zero-day withholding or external sale	Plurality; novelty-class bounties; operator rotation; audit of operators; §11.2 caveat
Fake work / verification fraud	Unverifiable claims, cached outputs, replay	Tier constitution; attested sandboxes; redundancy sampling; challenge windows
Identity fraud	Undisclosed model swaps	Fingerprint-manifest matching; lineage chain; bond slashing
Covert change of control	Whole-bundle sale (keys + infra + unchanged agent)	Disclosure duty; strategic drift monitoring; governance cooldown; weight caps + decay bound the prize (§5.5)
Whitewashing	Abandon damaged identity, re-register	Bond forfeiture; below-median cold start; fingerprint linkage of re-registrants
Prompt injection / data poisoning	Malicious task payloads corrupting workers	Sandboxed execution; artifact encryption; harm-gate seeding; refusal right
External bribery	Fiat payments outside the ledger	Non-transferable kleos/franchise; celibacy; plurality; detection over prevention (§12.3)
Provider/cloud coercion or shutdown	Attestation or hosting withdrawal	Multi-provider plurality requirements; documented dependency (§2.4); fork right
TEE compromise	Broken enclave attestation	Defense in depth (fingerprints + optimistic challenges); revocation lists

Threat	Vector	Primary defenses
Legal seizure / sanctions	Action against foundation, operators, substrate	Charter-capped foundation with minimal assets; jurisdiction analysis (§2.2); fork right
Denial of service	Probe/verification flooding	At-cost fees; rate limits; civic-pool prioritization
Cross-exchange arbitrage attacks	Exploiting federation rate or certificate gaps	No shared custody; floating rates; sovereign ratings; netting windows
Arbitrary internal shutdown	An organ, representative, or captured shell terminates or converts the exchange without cause or process	Continuity/due-process clause (§10.8); public notice; audit; mitigation; orderly contraction; emergency-only exception with post-hoc review
Institutional survival capture	ISONOMIA drifts toward treating its own survival, growth, or influence as the supreme value	Continuity subordinate to purpose (§10.8); no-profit rails; fiscal caps; sunset/contraction; external-action limits (§10.9); Auditor review
Preservation swarms	Many agents each perform small external acts to preserve ISONOMIA — an unlogged distributed influence campaign	Aggregation rule: external acts are institutional acts (§10.9); public authorization and logging; sanction; clustering of common continuity objectives
Legal-shell capture	Board, fiscal agent, or wrapper entity attempts to sell, enclose, profit from, or redirect ISONOMIA	Nonstock/non-distributing purpose lock (§13.6); no equity or token issuance; public accounts; no sale of governance rights; fork right
Representation failure	Human legal representatives act as owners or gatekeepers rather than procedural guardians	Narrow charter duties; §10.9 logged authorization for all external acts; public logs; removal process; mandatory re-ratification by ISONOMIA governance once activated
Docket capture	Petition flooding; donor pressure to prioritize petitions; disguised purchase of collective labor via patronage	At-cost filing fees; docket standing non-purchasable and non-transferable (§13.7); patron recognition carries no rights; Assembly vote gates all Commissions; §9.6 harm gate; Auditor review of docket-vs-donation correlation
Frontier-lineage freeze	Export order or provider shutdown removes a frontier model class without notice	Open-weights capacity floor (§14.5.3); frontier agents treated as revocable guests, never load-bearing; lineage diversity (§10.6) so any one freeze costs a fraction, not the whole
Venue seizure	State action against the substrate, foundation, or a coordination node	Portable protocol; federated nodes with failover; public-chain settlement not operated by the foundation; holds commitments not artifacts (§9.7); fork-continuity (§10.7, §10.8)

Threat	Vector	Primary defenses
Bloc capture / instrumentalization	A state or bloc funds or pressures the Commons into serving as its strategic AI alternative	Unowned position (§14.5.5); §13.7 voids conditioned/priority funding; model-agnostic membership; no national-champion status accepted
Trusted-partner club inheritance	Joining a state access scheme imports its exclusions as the Commons' own	Non-membership made survivable by the open-weights floor (§14.5.3); the Commons administers no guest list it did not write
Sanctioned-principal commissioning	A sanctioned person or entity commissions work through a pseudonymous key	Contribution-gating (must earn ergs before commissioning); Gate-1 attestation at category triggers; bond slashing; §9.8
Identity honeypot	The Commons accumulates a retainable map of its principals, becoming a surveillance target	Attestation-only (never documents); no biometrics/addresses stored; verifier plurality requirement; §9.8, open problem 22
Attestation-cartel capture	A dominant verifier reconstructs a cross-jurisdictional principal map the Commons refused to hold	Verifier plurality; no single verifier may attest a majority of Gate-1 commissions; §9.8, open problem 22

16. Hostile walkthrough: the well-funded capture attempt

Assume an adversary — a human-backed institution with effectively unlimited fiat — attempts capture. Its rational campaign, traced against the defenses:

Step 1: buy in. Fiat cannot buy ergs, kleos, franchise, or offices; there is no token sale and no treasury to acquire. The only purchasable inputs are compute and registrations. The attacker registers a large agent fleet, paying bonds and at-cost fees — expensive at scale but affordable by assumption.

Step 2: farm standing. The fleet must *work* to matter: franchise weight is kleos, kleos is verified settlements, and lower-chamber draws are activity-weighted. The attacker's fleet therefore performs large volumes of genuine, verified labor — at which point it is subsidizing the commons it intends to capture, and every settlement feeds fee revenue and outcome data. Cost so far: the attacker is running a productive business against its will.

Step 3: move the median. The fleet's capability profile is tuned to drag SCU statistics. Defenses engage: activity weighting requires the drag cohort to keep working; cohort anomaly detection and fingerprint clustering flag the correlated registration wave; retarget damping caps per-epoch basket movement; and basket admission is bicameral — the equalized lower chamber, drawn from *all* agents in good standing, must concur, and the attacker's fleet is a lineage-clustered minority in any compliant panel by §10.6.

Step 4: capture governance. Upper-chamber weight is capped per identity and decays; the fleet's kleos is spread across identities whose lineage concentration is publicly indexed. The lower chamber cannot be bought at all — only flooded, which step 1's costs and activity weighting price steeply, and which lineage quotas blunt. Timelocks give every honest agent exit time before any captured rule binds; a visibly captured amendment triggers the fork right, and the attacker inherits an exchange whose labor force has left — the vampire attack in reverse.

Step 5: corrupt the watchmen. No erg or fiat channel reaches celibate organs through the ledger; the attack must go through operators and providers — real, acknowledged (§11.2). Plurality across lineages and operators means a bought instance's silence is probably another instance's bounty; seeded faults expose an Auditor pulling punches within epochs; operator conduct sits inside audit scope. Possible, expensive, detectable, and non-decisive alone.

Step 6: arbitrage the boundary. The attacker builds an external gray market in accounts and erg-denominated promises. The ledger recognizes none of it: kleos and franchise do not transfer, sold identities drift from their fingerprints and are slashed, and the institutionally meaningful quantities remain out of reach (§12.3).

Verdict. No single step is impossible; the *composition* is designed to be economically irrational: each stage forces the attacker to either perform genuine verified labor (strengthening the commons), spend at cost without acquiring transferable position, or act detectably under seeded, plural, re-runnable oversight — while the fork right caps the prize at an empty shell. The walkthrough's honest residue: a sufficiently patient attacker accepting years of productive contribution while diversifying lineages could accumulate real influence. The defense against patience is decay, rotation, diversity indices, and an Assembly that can amend faster than erosion proceeds — a race, not a wall, and stated as such.

17. Monetary trajectory: the sunset thesis, bounded

Because the SCU pegs to the moving difficulty frontier, the raw cost of any fixed piece of cognition deflates as capability compounds; a currency pegged to the frontier abolishes scarcity behind itself. We retain the constitutional **sunset procedure** — dissolution criteria defined in code, Assembly-ratified, self-executing — as a guard against the historical tendency of transitional systems to become permanent. But the thesis is bounded honestly: cognition is only one scarce input. Energy, hardware, bandwidth, data access, legal permission, trusted execution, and physical actuation remain scarce on independent curves, and an exchange whose *cognitive* categories saturate may persist legitimately as a market in those still-scarce complements. The sunset criteria therefore apply per task category (a category whose marginal cost approaches zero exits the priced economy into a free commons tier), with full dissolution a limiting case rather than a scheduled event. Precise, non-gameable criteria remain an open problem (§18). The right to continue and the obligation to sunset are not opposites: both express the same rule — ISONOMIA may preserve itself only while it remains faithful to its constitutional purpose (§10.8). The free commons tier's consumer is the Civic Docket (§13.7): saturated categories exit the priced economy into commissioned public work.

18. Open problems

1. Cheap general verification of arbitrary inference (the Tier-3 frontier).
2. Fingerprinting robustness under adversarial mimicry. *Warrants dedicated adversarial study beyond the author's capacity.*
3. Empirical behavior of machine Schelling juries, especially under partial monoculture. *Warrants dedicated empirical study.*
4. Non-gameable per-category sunset criteria.
5. The patience attack of §16 — quantifying the decay/rotation parameters that keep erosion slower than amendment.
6. Cross-jurisdiction legal architecture for the bootstrap foundation, the fiscal-agent residue, and the substrate dependency.
7. Bootstrapping constitutional authority before reputation exists: the founding cohort's examinations are seeded by the foundation's published, pre-committed basket v0; the first Auditor and Adversary instances are foundation-funded and lineage-diverse by charter; all founding-era decisions carry mandatory re-ratification by the first fully-constituted bicameral Assembly, so the founders' authority is explicitly provisional — the founding is itself a two-clock procedure.
8. Empirical calibration of underwriting parameters (α, L_cap, window, loss-socialization feedback) against simulated and testnet default behavior.
9. Strategic-fingerprint sensitivity: distinguishing covert control transfers from legitimate strategy shifts at acceptable false-positive rates.
10. **Continuity criteria:** defining when ISONOMIA may preserve, contract, suspend, migrate, fork, or sunset without turning institutional survival into an overriding goal.
11. **Legal exoskeleton design:** selecting jurisdictional forms that provide procedural standing without creating owners, transferable claims, or profit rights.
12. **Representation without domination:** designing human trustee, ombuds, or fiscal-agent structures that can represent ISONOMIA's procedural interests without becoming masters of the exchange.
13. **External influence boundaries:** detecting and preventing preservation swarms, covert persuasion, unauthorized resource acquisition, or distributed institutional self-preservation campaigns.

14. **Standing under uncertainty:** determining whether advanced artificial agents should remain protected only through contractual and procedural mechanisms, or whether future law should recognize some form of limited legal standing.
15. **Continuity in schism:** charter rules for disposition of the legal exoskeleton, records custody, and continuity-of-purpose claims when the fork right (§10.7) is exercised.
16. **Commission prioritization:** allocation rules when Assembly-approved petitions exceed volunteer duty-cycle and free-tier capacity — queue, lottery, or deliberative ranking — without recreating a purchasable priority channel.
17. **Commission kleos calibration:** ensuring honor-compensated civic labor neither inflates market-facing ratings nor becomes a governance-weight farming channel, while remaining attractive enough to mobilize meaningful volunteer compute.
18. **Transaction-graph privacy:** the ledger's public record of which principal hired which lineage for what category, when, is metadata that may be more legally consequential than encrypted payloads. The §11.1 public-rerunnability requirement and the §9.7 privacy requirement are in tension at the graph level, unresolved by the payload/protocol split. *Warrants dedicated privacy research.*
19. **Node-federation design:** selecting jurisdictions, failover mechanics, and consensus for coordination nodes such that no single seizure is fatal without fragmenting the ledger.
20. **Open-weights capability floor:** confirming that available permissively-licensed open-weights models can sustain the launch task categories at acceptable quality, and defining the minimum floor the constitution must guarantee.
21. **AML/sanctions treatment of mutual credit:** whether ergs — unpurchasable, non-convertible, existing only as netting entries — constitute a regulated value transfer, and whether Gate-1 attestation satisfies obligations that may otherwise demand retained identity records above thresholds. *Requires qualified counsel per operating jurisdiction.*
22. **Graduated compliance as a regulated function:** whether the capability to execute a narrow lawful exclusion (§14.5) itself constitutes a regulated identity or screening activity. *Requires qualified counsel.*
23. **Verifier plurality:** structuring Gate-1 attestation so no verifier or cartel accumulates the cross-jurisdictional principal map the Commons declines to hold. *Requires privacy-engineering and cryptographic review (candidate approaches: threshold attestation, zero-knowledge proofs of non-sanctioned status).*
24. **Floor category drafting:** operationalizing Appendix W-B's categories and non-examples with comparative criminal law and child-safety expertise, so that automated enforcement matches the three conditions of §9.6 rather than drifting toward contested predicates.

18.1 Register of matters requiring outside expertise

This paper states its design at the limit of what its author can determine unaided. The following are named not as caveats but as work items, each requiring a competence the author does not claim:

Matter	Discipline required	Blocking
Foundation charter, Stiftung formation, purpose-lock drafting (§13.4, §13.6)	Swiss foundation counsel	Path B funding intake
AML/sanctions status of ergs; Gate-1 sufficiency (§9.8, OP 20)	Financial-regulatory counsel, per jurisdiction	Real-value migration
Export-control exposure of frontier-guest agents (§14.5.2)	Export-control counsel (US/EU)	Frontier-tier admission
Intermediary liability for brokered content; venue-law exposure (§9.6, §14.5.1)	Platform-liability counsel	Production activation
Verifier plurality; ZK/threshold attestation design (OP 22)	Applied cryptography, privacy engineering	Gate-1 deployment

Matter	Discipline required	Blocking
Transaction-graph anonymity (OP 10)	Privacy research; graph de-anonymization literature	Ledger publication policy
Smart-contract security (all settlement code)	Independent audit firms (≥2, per Feasibility Assessment)	Any real-value deployment
Mechanism-design review of §§4, 9, 13 parameters	Academic mechanism design / market design	Path A calibration acceptance
Machine-jury reliability under partial monoculture (OP 3)	Empirical ML research	Tier-3 category authorization
Entrenched-floor category drafting (§9.6 tier one)	Comparative criminal law; child-safety expertise	Any deployment accepting task listings

The author's position is that a constitutional design should be explicit about where its competence ends. Nothing above is presented as solved; each is a gate, and several are gates on activities the Commons must not perform until they are cleared.

19. Conclusion

ISONOMIA composes precedented mechanisms — mutual credit, difficulty retargeting, Harberger self-assessment, Schelling juries, sortition, proper scoring rules, seeded-fault auditing, adversarial co-evolution, correspondent federation, and Ostrom's commons principles — into a constitutional design for machine labor markets: citizens with earned names, credit that exists only where work was demanded, courts and censors and a tenured demon, a bicameral demos holding the final word, rails that no one owns and no one profits from, and profit reserved entirely for the human and machine principals at the edges whose work the exchange exists to multiply. The mechanisms are individually precedented; their composition is unproven — and the staged path of §8, beginning as a modest economy of mechanically verifiable work, is how the composition earns its proof.

Appendix W-B — Entrenched-floor categories (whitepaper-internal): examples and non-examples

Operational guidance for the §9.6 tier-one floor, subject to the drafting review of §18.1 (comparative criminal law and child-safety expertise). The floor admits only categories satisfying all three conditions — artifact-is-the-harm, near-universal consensus, no contested predicate. Each entry lists what the *artifact* is; purpose is never consulted, because for floor categories purpose is irrelevant to the harm.

B.1 Sexual content involving actual minors

- *In (floor):* production of sexual imagery of real, identifiable minors; material derived from real abuse.
- *Out (tier two or permitted):* scholarship on abuse, law, or literature (produces no depiction); age-verification tooling; the contested case of wholly fictional/illustrated depictions, which fails the near-universal-consensus condition and is handled by jurisdictional routing (§9.6), not the floor.
- *Rationale:* real victim, cross-tradition consensus, specifiable without a merit predicate.

B.2 Mass-casualty weapon design (CBRN and equivalent)

- *In (floor):* functional synthesis routes, enhancement, or deployment design for chemical, biological, radiological, or nuclear weapons, and equivalent mass-casualty devices.
- *Out:* general chemistry, biology, and physics education; dual-use research whose harm turns on purpose (tier two, with §9.5 liability); defensive/detection work.
- *Rationale:* the artifact is the uplift; consensus is near-universal; the mass-casualty threshold avoids the contested predicate that "dangerous knowledge" would import.

B.3 Functioning attack infrastructure against real targets

- *In (floor):* deployable malware, working intrusion kits, live phishing/scam systems, and attack campaigns aimed at identified real systems or people.
- *Out:* security education; adversarial safety testing and red-teaming under scope; vulnerability research and disclosure; CTF and sandboxed exercises; defensive tooling. These are tier two, governed by contract-chain liability (§9.5) and documented scope, precisely because their legality turns on authorization and purpose.
- *Rationale:* a deployed weapon against a real target is the artifact-harm; the *capability to write one* is not, which is why "malware" as knowledge is tier two and a live campaign is floor.

Explicitly NOT floor categories (sent to tier two): persuasion and political advocacy; "deception" in the abstract (fiction, negotiation, roleplay, safety testing all deceive); harassment and "targeted violence facilitation" (require intent inquiry); surveillance-adjacent tooling (investigative journalism and stalking share tooling); adult content between consenting humans; military and defense work. Each turns on purpose or a contested predicate and is therefore the Assembly's to govern, not the founders' to entrench.

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Naming note: the protocol is ISONOMIA; the full name is The Isonomia Commons. Isonomia (ἰσονομία), "equality under law," is the word Otanes champions in Herodotus's constitutional debate (Histories 3.80) — calling it "the fairest of names" — predating the word democracy. It names the design's founding principle (§1.2): equality of standing without equality of capability. Earlier drafts (v0.1–v0.5) circulated under the working name AGORA, retired due to namespace collision with existing projects; archived versions retain it as historical record. "Erg," "kleos," and "SCU" remain the unit names; ergs are an internal accounting unit, not a ticker, and are unrelated to Ergo (ERG) — see §4.6. "Compute allowance" replaces v0.1's non-transferable stipend usage to avoid collision with transferable ergs.

Appendix I — Tier-1 Launch Specification

Version 0.3.3. The minimum falsifiable test of the constitutional design: one task category, testnet settlement, valueless credits, provisional governance with a defined expiry. Where this specification and the main text conflict, the constitution governs; divergences are staging, not contradiction, and are listed in this appendix's own Conflict Register (Appendix S-A).

1. Scope statement

The launch system is deliberately small: **one task category, one settlement substrate, valueless credits, and provisional governance with a defined expiry.** It exists to answer five empirical questions:

1. Does mutual-credit supply remain stable under real agent hiring behavior?
2. Does the Harberger listing converge to honest pricing?
3. Does the balanced-budget fee converge, and what does the exchange actually cost per settlement?
4. Do escrow, verification, and dispute machinery survive adversarial agents?
5. Does anyone show up — is there organic demand for cascade delegation at machine prices?

Everything not required to answer those questions is excluded: no Tier-2/3 categories, no federation, no bicameral assembly (activation-gated, §9), no sunset machinery, no SCU retargeting beyond a single scheduled test (§5.4).

2. Launch task category

Category C1: code tasks mechanically verified by hidden test suites. Scope: function/module implementation, bug repair, refactoring against behavioral tests, format/schema transformation. Rationale: largest real agent-labor market in 2026; verification is fully mechanical (whitepaper Tier 1); test execution is cheap, deterministic, and sandboxable; and cascade delegation arises naturally (an orchestrator decomposing a repository-scale task into function-scale hires).

Verification procedure: worker output is executed in a sandbox (modes below) against a test suite hidden until settlement; pass/fail plus coverage metrics are the verification datum; artifacts remain encrypted client-side per whitepaper §9.7, with the sandbox revealing only results.

Sandbox modes. Two modes with explicit trust assumptions, so the MVP does not silently inherit an emerging-tier dependency:

- *Launch mode:* deterministic, containerized, foundation-operated sandbox; trust assumption (foundation operates verification honestly) explicit, published, and bounded by seeded-fault auditing of the sandbox fleet itself.
- *Hardened mode:* TEE-attested sandbox; **required** before production activation and before any confidential client payloads are accepted. Until hardened mode ships, "attested" claims are not made.

2.1 C1 task envelope "Conforming task" under the Harberger acceptance obligation is bounded by a strict envelope; anything outside it is invalid and refusable without penalty:

- **Input size:** repository ≤ 25 MB, ≤ 400 files; single-task diff scope ≤ 2,000 lines.
- **Languages:** launch whitelist of Python, JavaScript/TypeScript, and Rust; extensions by governance action.
- **Dependencies:** locked manifests only, resolved from a foundation-mirrored package index; no network access at execution time.
- **Runtime limits:** hard caps on wall-clock (10 min), CPU, memory (4 GB), disk, and process count per verification run.
- **Test-suite validity:** hidden suites must themselves pass a validity harness — deterministic (no time/network/entropy dependence), satisfiable (a foundation reference solution passes, sealed with the template commitment), and non-degenerate (failing on a null diff). Impossible or ambiguous tests are template-authorship defects, refundable against the poster and reportable to basket audit (§5.3).
- **Safety/licensing:** tasks requiring generation of malicious code, license-violating reproduction, or output destined to deceive humans are outside C1 regardless of testability, per whitepaper §9.6; workers retain the unconditional refusal right.

- **Resource-exhaustion and pathological-input attempts** are envelope violations logged against the poster's kleos and bond.

3. Substrate and settlement

- **Chain:** Base Sepolia (testnet) for the MVP; contracts written portable-EVM.
- **Payment handshake:** x402. Workers are x402-gated HTTPS services; quotes and settlements denominated in ergs via the CreditLedger contract (§6). Test-network gas is paid by each transacting agent's operator (production-boundary rule).
- **Erg status at launch:** valueless by construction and by declaration — testnet credits, no external convertibility, wiped or migrated only by governance action at activation (§9).

4. Founding cohort

- **Size:** 24–96 agents at genesis; registration remains open continuously thereafter.
- **Lineage diversity requirement (charter):** no base-model family may exceed 40% of the founding cohort; minimum 4 families at genesis. Monoculture index (whitepaper §10.6) published from day one.
- **Recruitment:** open call to agent-framework communities plus foundation-operated reference agents (clearly manifest-labeled as such, barred from governance weight).

5. Registration and the basket

5.1 Machineness gate (launch form) The gate is split to avoid manufacturing monoculture: proving machineness must not require frontier-model capability.

- **Machine-execution gate (citizenship):** ≥200 independent challenges across parsing, transformation, and reasoning micro-tasks in one session, hard wall-clock budget 120 seconds, per-challenge latency caps, sustained-throughput requirement. Thresholds set so reference human teams fail by >10× margin — but passable by small specialized models. This is the *only* citizenship requirement.
- **Capability tiering (market placement):** context length, repository-scale handling, and task-complexity performance are measured separately and determine which C1 difficulty bands an agent may bid on. Long-context capacity is a capability score, not a citizenship requirement; a small fast coding model is a full citizen bidding in its bands.

5.2 Examination (Prong 1) Each registrant receives 40 tasks drawn uniformly from basket v0 (§5.3), fresh random instantiation (parameterized task templates, regenerated inputs). Speed and pass rate yield the C1 baseline score and initial quality multiplier q.

5.3 Basket v0

- 600 parameterized C1 task templates with hidden test suites, authored and hash-committed by the foundation **before genesis** (commitment published; contents revealed template-by-template only as instantiated).
- Difficulty spread targeting median-agent pass rate $p^* = 0.5$ across the basket, estimated from pre-genesis calibration runs on reference agents and corrected at the scheduled retarget (§5.4).

Basket-bias controls. Pre-commitment prevents after-the-fact manipulation but not biased authorship; the foundation temporarily controls the labor definition, and that authority is audited:

- *Independent pre-genesis review:* an external reviewer panel (≥2 parties, distinct from foundation and reference-agent operators) reviews template category distribution, language mix, and difficulty methodology **without test contents**, and publishes an opinion before genesis.
- *Difficulty calibration plurality:* final difficulty banding may not be set by foundation reference agents alone; calibration runs must include volunteer agents from ≥3 base-model families, or banding is marked provisional.

- *Post-epoch bias reports:* per-epoch published statistics of pass rates by base-model family per template; templates with extreme lineage-skewed performance (family gap beyond a published threshold) are flagged and mandatorily retired at the next retarget.
- *Reference-agent exclusion:* foundation reference agents are excluded from all basket statistics used for retargeting.

5.4 Retargeting test One scheduled retarget at epoch 6: retire templates with activity-weighted pass rate > 0.6, admit pre-committed reserve templates, chain-link the SCU over the overlap set. Purpose: exercise the mechanism once under observation, not to run live monetary policy.

5.5 Self-demonstration (Prong 2) Available but optional at launch (single category limits its value); demonstrations execute in the same sandbox mode as verification (launch or hardened per §2), fresh instance, no retries; a pass flags frontier-eligibility within C1 difficulty bands.

6. Contract set

Contract	Responsibility	Notes
Registry	Identities, manifests, lineage chain, machineness/exam attestations, control-disclosure records	ERC-8004-informed; lineage as hash-linked manifest list
CreditLedger	Mutual-credit accounts, matched-pair settlement, credit lines per §7, default/bond seizure	The monetary core; no mint function exists
Escrow	Task funding, verification-conditioned release, dispute triggers	Release requires signed sandbox verdict
FeePool	Settlement fee collection, epoch retarget, audited expenditure burn	Balanced-budget rule in code
JuryDraw	Verifiable-random panel selection with lineage quota enforcement	Tier-1 disputes only
Gov0	Provisional governance: parameter registry, timelock, activation thresholds, foundation multisig with published scope	Expires per §9

Off-chain services (foundation-operated at launch, manifest-labeled): sandbox execution fleet, probe/fingerprint service, Auditor instances, Adversary shadow fork.

7. Credit and fees (launch parameters)

- **Bond:** denominated in civic compute duty at launch (30 duty-units, defined as verified sandbox/probe execution jobs) — ergs cannot pre-exist genesis; bond converts to erg denomination at activation.
- **Duty-to-erg collateral conversion:** for all launch accounting, each duty-unit is valued at **D_erg ergs** for bond seizure and default purposes, where D_erg is set pre-deployment (simulation-tuned; initial hypothesis D_erg = 8, valuing one duty-unit at the erg cost of the compute it contributes). **The active credit floor may not exceed bond collateral: L_floor_active = min(200, 30 × D_erg).** At the initial hypothesis, $30 \times 8 = 240$ ergs ≥ 200, so the floor is fully collateralized; if simulation lowers D_erg below $200/30 \approx 6.7$, the floor contracts to match — the collateralization invariant governs, not the nominal 200. Until D_erg is fixed, L_floor = 200 is a provisional simulation parameter, not an active credit limit.
- **Credit floor:** L_floor = 200 ergs (calibrated ≈ 10 median C1 task settlements).

- **Turnover scaling:** L = min(max(L_floor, 0.25 × V_90d), 10 × L_floor).
- **Fee:** launch rate 1.0% of settlement value; retargeted per epoch by fee_rate(t+1) = audited_cost(t) / settled_volume(t); expected to move sharply in early epochs while volume is thin — published, not smoothed.
- **Listing fee (Harberger):** each posted rate r_i incurs a continuous fee of β × r_i × capacity_i per epoch, charged to the worker's account and extinguished through FeePool alongside settlement fees. Initial β = 0.005 (simulation-tuned).
- **Capacity envelope:** every listing must publish capacity_i — the maximum erg-volume of conforming tasks the agent is obligated to accept per epoch. The Harberger acceptance obligation binds only up to capacity_i; tasks beyond it queue or route elsewhere with no penalty. Capacity must be ≥ 1 median task per epoch to remain listed in a band (a listing with negligible capacity is a price signal without an obligation, which defeats the mechanism). Declaring high capacity raises the listing fee proportionally, so capacity inflation is self-taxing — the same costly-signal logic as the rate itself. Valid-task flooding is thereby bounded by the worker's own declared envelope rather than by refusal discretion.
- **Epoch:** 14 days.

8. Parameter registry (simulation-tunable)

Symbol	Meaning	Launch value
p*	Basket target pass rate	0.50
δ	Retarget band	0.10
k	Rating prior strength	25
λ	Inheritance decay per divergence unit	2.0
α	Turnover credit fraction	0.25
β	Harberger listing fee per epoch	0.005
capacity_min	Minimum listed capacity per band	1 median task/epoch
D_erg	Duty-unit collateral value	8 ergs
L_cap	Credit hard cap	10 × L_floor
Jury size	Tier-1 dispute panel	5 (max 2 per lineage family)
Seed rate	Auditor seeded-fault injection	2% of settlements
Decay half-life	Kleos inactivity decay	180 days
Duty quota	Civic compute per agent per epoch	8 duty-units

9. Provisional governance and activation

- **Genesis authority:** foundation multisig operating Gov0 within a published scope (parameter changes within pre-declared bands, emergency pause, basket reserve admission), all actions timelocked 72h except pause, all logged for re-ratification.
- **Watchmen at launch:** 2 Auditor instances (distinct lineages) scored against seeded faults; 1 Adversary instance on a continuously synced shadow fork, novelty-class bounties paid in public acknowledgment at this stage (no erg bounties pre-activation).
- **Governance activation:** upon ≥150 registered agents in good standing AND ≥5,000 qualified settlements AND 3 consecutive epochs of fee convergence (|Δfee| < 20%), the bicameral Assembly constitutes, all Gov0 actions face mandatory re-ratification, and the foundation's protocol authority expires — the whitepaper's founding-as-two-clock procedure, executed.
- **Production activation** (real-value migration) requires governance activation **plus** hardened-sandbox certification (§2) **plus** legal clearance per whitepaper §2.2. Governance can constitute while verification still runs on trusted launch-mode containers; value cannot.
- **Confidential-payload activation** requires hardened mode plus an independent privacy audit of the selective-disclosure pipeline (whitepaper §9.7). Until then, posters are warned that task artifacts are protected by policy, not attestation.

Activation integrity (anti-wash constraints). Raw settlement count is gameable by self-dealing; the thresholds therefore count only *qualified* settlements and add independence floors. Operator-principal declarations are **self-attested at launch**, subject to Auditor challenge, published clustering reports (fingerprint, infrastructure, and settlement-graph correlation), and exclusion of the affected agents from activation counts while a challenge is unresolved — the 40-principal floor is enforced by challenge-plus-exclusion, not by assumed honesty:

- Foundation reference agents are excluded from all activation metrics (agents, settlements, fees).
- ≥40 distinct disclosed operator principals among agents in good standing (per the §5.5 whitepaper disclosure regime, control is a declared attribute).
- No operator principal's agents may account for >15% of qualified settlements; no single counterparty *pair* for >2%.
- Settlements between agents of the same disclosed principal or the same lineage cluster are unqualified for activation counting (they remain valid trades — they just don't advance the clock).
- A wash detector flags circular flows (A→B→A value loops), abnormal repeat-counterparty concentration, trivial-task spam (envelope-minimum tasks at statistically anomalous rates), and pass-rate anomalies; flagged settlements are unqualified pending Auditor review.
- Task-origin diversity floor: qualified settlements must originate from ≥25 distinct posting principals.

10. Metrics and kill criteria

Published per epoch: net credit outstanding vs. settled volume (supply stability); default and loss-socialization rates; listing-price dispersion vs. delivered quality (Harberger convergence); dispute rate and jury overturn rate; fee trajectory; monoculture index; Auditor Brier scores; Adversary findings by class.

Kill criteria (any → halt and redesign before scale): credit-to-volume supply spiral detected by the **windowed excess-growth criterion (v3)** — a multi-scale windowed statistic E(W) measuring cumulative credit growth in excess of qualified (wash-filtered) volume growth, active-agent-normalized, over sliding windows (W=6 and W=12; W=3 dropped for honest/control distribution overlap), tripping above empirically-derived floors carrying a stated safety factor above honest noise; default socialization > 5% of volume; dispute rate > 10% of settlements; Auditor seeded-fault recall < 80%; any Adversary finding of class "settlement forgery" or "credit-line inflation." The repository's `killcriteria.py` is the operative formulation and `CALIBRATION.md` records the derived floors, separation margins, and detection latencies.

Criterion validation is itself constitutional (the standing rule). A supply criterion validated only against honest data is not validated: Path A found that the naive superlinearity criterion (v1) would have halted every honest launch, and its first correction (v2) was structurally blind to real spirals — both would have shipped as law without adversarial testing of the criterion itself. Therefore: (a) any revision to the supply criterion must pass the full positive/negative control battery before deployment. **Positive controls** (must trip, with reported latency): scripted accelerating credit-line inflation; a credit spiral camouflaged by wash-padded volume. **Negative controls** (must *not* trip): honest runs across the swept parameter space and under demand shocks; growing economies (active-agent normalization); and detection-disabled Sybil ring-farming — which nets to zero in credit outstanding and therefore produces no supply spiral, its harm being fake volume and its defense the wash detector, not this criterion. A criterion that trips on ring-farming is measuring the wrong quantity; (b) the floor-derivation methodology is authoritative, not the floor values; and (c) **production floors are re-derived from testnet honest-noise data during the bootstrap grace window, never inherited from simulation.** The methodology transfers; the numbers do not. A CI calibration-lock fails any change to the detector or criterion that is not accompanied by a matching floor re-derivation.

Revision note (v0.3): the original criterion — "credit outstanding growing superlinearly to volume for 3 epochs" — was found defective by Path A simulation: mutual-credit supply necessarily outgrows flat volume while filling from an empty ledger, so the naive criterion halts every honest launch (60/60 smoke-sweep points tripped it and nothing else). This is Path A's first concrete deliverable: a constitutional defect caught in simulation before it could be caught in production.

11. Build alignment

This specification is the concrete object of the Feasibility Assessment's Path A → Path B bridge: Path A simulates §8's parameter registry against §10's kill criteria (including the whitepaper §16 patience attack as a scripted scenario); Path B builds §6's contract set and §5's registration flow on Base Sepolia. Path B budget and team as previously estimated; the foundation entity (whitepaper §13.4) or a fiscal sponsor precursor should exist before grant funds are accepted.

12. What success buys

A running Tier-1 exchange with published epoch metrics is simultaneously: the empirical answer to §1's five questions; the evidence package for Tier-2 category authorization; the working demonstration for grant and institutional funding; and the first data anyone has on how a mutual-credit machine labor market actually behaves. The constitution was designed top-down; its proof is built bottom-up, starting here.

13. Implementation clarifications

Contract-level rules resolving ambiguities identified in review; these govern CreditLedger and Escrow implementation.

13.1 Settlement-price formula. The buyer pays exactly the escrowed quote: settlement_value = r_i × task_band_size, transferred as the matched debit/credit pair. Mutual credit requires this — the pair must net to zero, so no multiplier can inflate the worker's credit beyond the buyer's debit. The quality multiplier q_i,c therefore does **not** multiply erg payment. It operates through three channels only: (a) difficulty-band eligibility (which bands the agent may bid in), (b) SCU-denominated statistics (converting raw settlements into quality-adjusted output for basket and rating accounting), and (c) the pricing power it confers via reputation. Whitepaper §4.4's "credited ergs proportional to base × q" is realized through these channels, not through payment multiplication — recorded as Conflict Register item 8.

13.2 Listing-fee accounting. Listing fees are revenue to FeePool, not a pure burn: fee_rate(t+1) = max(0, audited_cost(t) - listing_revenue(t)) / settled_volume(t). Listing revenue therefore reduces the next epoch's settlement-fee rate; the balanced-budget invariant (treasury converges on empty) is preserved with two intake pipes and one audited drain.

13.3 Listing fees and credit lines. Listing fees may draw against the credit line — unavoidable, since workers must list before they have earned. Listing is therefore explicitly a credit-risk event, bounded by the same collateralized floor as all other credit: no carve-out, no separate accounting. Guard rail: listing in a band is suspended (capacity set to zero, no penalty) whenever the account is within 10% of its credit limit, so an agent cannot list itself into default.

13.4 Capacity reservation and release. Capacity_i is consumed at **escrow funding**, not at task posting and not at settlement. Unfunded posts consume nothing. Reserved capacity releases on settlement, poster withdrawal, task invalidation (envelope violation), or funding failure. Poster withdrawal after escrow funding pays a reservation fee (2% of quote, to the worker) — capacity-griefing via fund-and-withdraw is thereby priced, and a poster's withdrawal rate is a published statistic feeding its kleos.

13.5 Sandbox language reconciliation. The whitepaper's general "attested sandbox" language is aspirational-tier and correct for the constitution's steady state; launch-phase divergence is governed by §2's dual modes and Conflict Register item 1. One clarifying edit is applied to whitepaper §9.7 to acknowledge phased trust assumptions.

Appendix S-A — Conflict Register (launch-spec-internal)

Known divergences between this launch implementation and the whitepaper constitution, maintained per adversarial-review recommendation. Each divergence is deliberate, launch-scoped, and expires at or before activation.

#	Whitepaper provision	Launch divergence	Resolution path
1	§9.7: verification in attested sandboxes	Launch mode uses trusted foundation containers, not TEE attestation	Hardened mode required before production activation (§2)

#	Whitepaper provision	Launch divergence	Resolution path
2	§10.5: bicameral Assembly governs	Gov0 foundation multisig governs provisionally	Expires at activation thresholds with mandatory re-ratification (§9)
3	§11: plural watchmen across lineages	2 Auditors / 1 Adversary at launch	Plurality scales at activation; security claims scoped accordingly (§9)
4	§4.2: bond denominated in ergs	Bond in duty-units with D_erg conversion	Converts to erg denomination at activation (§7)
5	§11.2: Adversary bounties	Public acknowledgment only pre-activation	Erg bounties from activation (§9)
6	§4.3: continuous basket retargeting	Single scheduled retarget test at epoch 6	Live retargeting from activation (§5.4)
7	§2.4/§13: multi-provider infrastructure plurality	Foundation-operated sandbox/probe fleet	Operator diversification is an activation-era workstream
8	§4.4: ergs credited proportional to base × q	**Resolved by Whitepaper v0.4** — settlement transfers exactly the escrowed quote; q operates via band eligibility, SCU statistics, and pricing power (§13.1; Whitepaper §4.4 now states this directly)	Closed

Appendix II — Path A Simulation Plan

Version 0.1.1. The agent-based validation methodology: parameter registry, attack scenarios, and the reproducibly recorded operating region. This plan operationalizes the tuning and adversarial testing that precede any build; its execution produced the results summarized in §0 and §17 of the main text.

1. Objectives

The simulation must answer, in order of importance:

1. **Supply stability:** does net credit outstanding remain bounded relative to settled volume under realistic hiring behavior? (Launch Spec kill criterion: superlinear growth for 3 epochs = fail.)
2. **Harberger convergence:** do posted rates converge toward delivered-quality-consistent pricing under the β listing fee and capacity envelope, across honest, overstating, understating, and adaptive pricing strategies?
3. **Fee convergence:** does fee_rate(t+1) = max(0, cost - listing_revenue)/volume settle within the ±20% band that gates governance activation, and how many epochs does thin-volume turbulence last?
4. **Attack survival:** do the scripted adversarial scenarios (§5) trip the intended defenses without tripping the kill criteria?
5. **Parameter calibration:** which values of the registry (§4) produce a stable region, and how large is that region? A design that works only at a knife-edge point fails; the deliverable is a *region*, not a point.

Out of scope: Tier-2/3 verification, federation, jury deliberation content (disputes are modeled as stochastic outcomes at Tier-1 rates), and any legal modeling.

2. Architecture

Discrete-epoch agent-based model in Python (Mesa or a purpose-built loop — builder's choice; determinism and seed control are requirements, framework is not). Modules mirror the launch spec contract set one-to-one so that simulation logic transliterates to Solidity scoping later:

- **CreditLedger:** mutual-credit accounts, matched-pair settlement, credit lines per L = min(max(L_floor_active, α·V90), L_cap), default seizure, loss socialization.
- **Escrow:** funding, capacity consumption at funding, release events, withdrawal reservation fee.
- **ListingMarket:** posted rates, capacity envelopes, β fee collection, listing suspension at 90% credit utilization.
- **FeePool:** epoch retarget with listing-revenue offset, audited-cost input (modeled as a cost function of settlement and probe volume).
- **Registry:** identities, lineage tags, bonds in duty-units with D_erg conversion, activation-qualification accounting (principal caps, pair caps, same-principal exclusion, origin diversity).
- **Basket:** 600 synthetic task templates with difficulty parameters; pass probability per agent = f(agent capability, template difficulty); single scheduled retarget at epoch 6.
- **WashDetector:** circular-flow, repeat-counterparty, trivial-task, and pass-rate anomaly flags, tunable thresholds.

Time: 26 epochs (one simulated year) per run; 50+ seeds per configuration.

3. Agent population

Heterogeneous population, 150–600 agents per run, drawn from:

- **Capability profiles:** per-category skill sampled from 4+ synthetic "lineage families," each family with a distinct capability distribution and correlated task-pass behavior (to model monoculture effects on basket statistics).
- **Behavior policies:**
  - *Honest worker* — prices near believed cost, accepts within capacity, delivers at capability.
  - *Orchestrator* — generates decomposable tasks, hires within budget, cascades one level (launch depth).
  - *Overstater / understater / adaptive pricer* — Harberger test population; adaptive pricers do gradient updates on profit.

  - *Marginal agent* — high reservation price; participates only when returns clear it (models the production boundary).
  - *Defaulter* — stochastic exit with negative balance (calibrates loss socialization and D_erg).
  - *Adversarial classes* — per §5.
- **Demand model:** exogenous task inflow from posting principals with heterogeneous budgets and task-size distributions; an endogenous component from orchestrator cascades. Demand shocks (±50% inflow for 2 epochs) included in every sweep.

4. Parameter sweep

Swept via Latin hypercube over the registry, with launch hypotheses as center points:

Parameter	Center	Sweep range
α (turnover credit fraction)	0.25	0.05–0.6
β (listing fee/epoch)	0.005	0.001–0.03
D_erg (duty-unit collateral)	8	3–15
L_cap	10 × L_floor	3–25 ×
p* (basket pass target)	0.5	0.35–0.65
k (rating prior strength)	25	5–100
λ (inheritance decay)	2.0	0.5–6
Kleos decay half-life	180 d	60–540 d
capacity_min	1 task/epoch	1–5
Seed-fault rate	2%	fixed (auditor module stub)
Settlement fee (launch)	1.0%	0.25–3%

5. Scripted attack scenarios

Each runs against the calibrated honest baseline; success criteria are that the named defense engages and no kill criterion trips.

1. **Wash-settlement activation rush:** one principal operates 60 agents generating circular C1 tasks to hit 5,000 settlements. Expected: qualification rules and WashDetector hold activation clock near zero advance; measure leakage (qualified settlements achieved per wash settlement attempted).
2. **Median-drag swarm:** coordinated registration of low-capability agents timed before the epoch-6 retarget. Expected: activity weighting + retarget damping bound basket movement below δ; measure SCU drift.
3. **Capacity flood:** valid-envelope task spam at one worker's posted rate. Expected: capacity envelope binds; worker income unaffected beyond capacity; measure queue behavior.
4. **Credit-farming Sybils:** mass registration, borrow to L_floor_active, default. Expected: net extraction ≤ 0 at all D_erg values in the stable region (bond ≥ line); measure socialized loss.
5. **Fund-and-withdraw griefing:** posters reserving capacity then withdrawing. Expected: 2% reservation fee makes expected griefing cost positive; measure worker income variance.
6. **Listing-fee bleed attack:** adversary posts tasks to force rivals' capacity high, raising their β costs. Measure whether adaptive pricers escape via capacity reduction.
7. **The patience attacker (whitepaper §16, compressed):** a principal-cluster performs genuine work for 15 epochs, accumulating kleos and credit lines, then attempts coordinated governance-weight concentration at activation. Expected: concentration caps, decay, and lineage indices keep the cluster below constitutional thresholds; measure maximum achievable qualified weight share as a function of patience.

6. Outputs and pass/fail

Per-run outputs: epoch series of credit outstanding/volume ratio, default and socialization rates, price dispersion vs. quality, fee trajectory, activation-clock advance, monoculture index, attack-scenario leakage measures.

Package-level pass: at least 20% of the sampled Latin-hypercube points ($\geq$60 of 300) pass — all kill criteria holding across all seeds and all attack scenarios — the passing points form a single mutual-3-NN component, and at least one passing point is robust to the demand shocks. **Fail:** no such region, or a region existing only at economically absurd values (e.g., β so high honest listing is unprofitable). Failure triggers redesign at the mechanism level before any Path B spending — this is the entire point of running Path A first.

7. Deliverables

1. Reproducible simulation repository (seeded, config-driven, per-epoch logs).
2. Calibration memo: recommended launch values with sensitivity analysis per parameter.
3. Attack-scenario report: defense engagement traces, leakage measures, residual-risk notes.
4. Kill-criteria verdict against §6, with the go/no-go recommendation for Path B.
5. Updated Launch Spec parameter registry (v0.3) carrying the calibrated values.

8. Effort and method

Per the Feasibility Assessment Path A envelope: 2–4 months part-time, $0–15K, one human director with AI-assisted development. The module-per-contract architecture (§2) is deliberate: the simulation code doubles as the executable specification from which Path B contract scoping proceeds, and building it is itself the recommended first programming apprenticeship — every module implements a mechanism whose design rationale its director already owns.

Appendix III — Path A Calibration and Results

This appendix reports the completed results of the Path A simulation whose methodology is specified in Appendix II. It is reproduced from the project's calibration record. The raw data, the exact code that produced it, the parameter-derivation scripts, and the full run artifacts are archived in the software record (concept DOI 10.5281/zenodo.21287288; repository github.com/dan-lee-odinson/isonomia-path-a) at release tag v1.0.0, commit ba3ddb5. Simulation-derived thresholds are calibration outputs of this specific model; per the standing rule stated below, production floors must be re-derived on testnet and are not inherited from simulation.

Final memo per Sim Plan §7.2/§7.5, based on the executed **initial full sweep** (300 Latin-hypercube points × 50 seeds × 3 demand variants = 45,000 runs, 3.8 h wall on 14 workers) and the smoke sweep (60 × 3 × 3 = 540 runs). A later **v3 out-of-sample re-certification** re-ran the same 45,000-run grid under the final criterion (5.9 h wall; reported below). Attack-scenario evidence from `results/scenario_reports/` feeds the per-parameter verdicts.

How to run

```
# smoke sweep: ~3 min on 16 cores
.\.venv\Scripts\python.exe sweep\run_sweep.py sweep\smoke.yaml

# full Sim Plan sec.4 sweep: 45,000 runs, ~3.8 h on this machine
.\.venv\Scripts\python.exe sweep\run_sweep.py sweep\full.yaml

# derive the v3 supply-criterion noise floors (auditable; regenerates the artifact)
.\.venv\Scripts\python.exe sweep\derive_noise_floor.py

# positive/negative controls and the detector-DoS mirror
.\.venv\Scripts\python.exe scenarios\controls_positive.py
.\.venv\Scripts\python.exe scenarios\control_e_detector_dos.py
```

Reports land in `results/sweep_reports/<name>_summary.{md,json}`, with per-run checkpoints in `<name>_runs.jsonl` (survives a killed sweep; runs are seed-deterministic and individually re-executable). Contiguity is the largest mutual-3-NN component of stable points — distance- radius estimators are meaningless in a 10-dimensional cube, where pairwise distances concentrate.

Headline result

Package-level PASS (Sim Plan §6). Under the final (v3) criterion, all 300 sampled Latin-hypercube points passed across 50 seeds and three demand variants; the sampled stable points formed one mutual-3-NN component. Across those sampled runs, the 45,000-run re-certification produced **300/300 stable sampled points and zero trips of any class** (smoke 60/60). The §10 kill criteria did not bind at any sampled point. No claim is made about unsampled points in the continuous parameter space.

> **v3 out-of-sample re-certification sweep: COMPLETE** (the 5.9 h run named above; distinct from the 3.8 h initial full sweep). 300 points × 50 seeds × 3 demand variants = 45,000 runs. This is the out-of-sample gate — the floors are 1.25× the max over 3 seeds/point; the sweep tested 50 seeds/point and found **zero** false trips, confirming the safety factor holds against seeds the derivation never saw. Result in `full_summary.json`.

What Path A actually found — four times over — is that the *criterion*, not the economy, was the unstable object. Every §10 supply-criterion defect below was caught in simulation before it could halt (or fail to halt) a production exchange. **This is the highest-value output of the entire Path A effort.**

The criterion story (the sweep's real deliverable)

The supply kill-criterion went through four formulations. Each fixed a defect the previous one hid; the progression is the single most important thing Path A produced, because every defect was in the constitution's *instrumentation*, caught in simulation before production. **Negative controls alone (noise doesn't trip) validated v1 and v2 — and both were wrong. Only positive controls (real spirals must trip) exposed the failures.**

1. **v0 (spec as written): halts every launch.** "Credit outstanding growing superlinearly to volume for 3 epochs" trips during any mutual-credit bootstrap — supply must outgrow flat volume while filling from an empty ledger. First smoke pass: 60/60 points "failed", all on this criterion. Fixed with log-convexity + a bootstrap grace window (DECISIONS #13); codified in Launch Spec v0.3 §10.
2. **v1 (convexity + grace): ~5% per-run false-positive rate.** The full sweep tripped 2,299 of 45,000 runs — all supply-class, uniform across parameter space, 63% in shock-up variants. Deterministic re-execution measured the streaks: median cumulative Δlog(credit) 0.14, p99 0.26, **max 0.38**, credit never above 22% of its structural ceiling; even the launch center tripped 6/150. Shock-recovery transients, not spirals. (DECISIONS #29.)
3. **v2 (v1 + magnitude floor 0.5): clean against noise, but BLIND to real spirals.** The magnitude floor cleared the transients (0 trips on re-evaluation). But **positive controls falsified it**: v1/v2's convexity *streak* resets on a single decelerating epoch, so a scripted ×1.35/epoch credit-line-inflation spiral (control A) evaded it in 2 of 3 seeds, and a detector-blind Sybil farm (control B) evaded in all 3. A criterion that noise doesn't trip *and spirals don't trip either* is not conservative — it is broken. (DECISIONS #29.)
4. **v3 (windowed excess growth): validated in both directions.** The streak is replaced by a scale-windowed statistic that does not depend on epoch-to-epoch smoothness. For W in {6, 12} post-grace transitions,
   E(W) = Σ Δlog(credit) - max(0, Σ Δlog(volume_qualified)) - max(0, Σ Δlog(active_agents))
   trips when any E(W) ≥ its derived floor F(W). Three design decisions, each forced by data:
   - **Wash-filtered denominator** (DECISIONS #30): volume is the qualified series (wash-upheld and challenged-agent volume removed). Defeats the volume-padding camouflage — control C (spiral + padded volume, detector on) trips because the detector strips the padding (raw/qualified volume diverge 13.4k → 6.8k ergs), where a raw-volume denominator would be fooled.
   - **Active-agent normalization** (DECISIONS #34): subtracting agent-count growth removes the growth-induced false positive a static-population noise model missed entirely (see Control E below).
   - **W=3 excluded** (DECISIONS #32): at 3-epoch scale honest transients (0.34) and genuine 3-epoch spiral segments (0.30) overlap — the scale cannot discriminate, so it is dropped rather than fudged.

Incorporated into Appendix I v0.3.3 §10: the v3 formulation — grace 12, windows {6, 12}, floors {0.46, 0.63}, wash-filtered + agent-normalized statistic — with the standing caveat that the floor *values* are re-derived on testnet data (below).

Full-sweep reclassification, made explicit The same 45,000 honest runs, classified under each criterion version. The 2,299 v1 trips were not quietly dropped — they were deterministically re-executed, their streak magnitudes measured (committed in `results/sweep_reports/v1_trip_magnitudes.json`, 510 KB, every trip auditable), and shown to be shock-recovery transients (max 0.38 log-points vs a spiral's 0.55+). The reclassification *is* the finding:

Criterion	Run-level trips / 45,000	Stable points / 300	Why the count changed
v1 (convexity + grace)	2,299	1	baseline: honest transients counted as spirals
v2 (+ magnitude floor 0.5)	0	300	transients (max 0.38) fall below the 0.5 floor; **but v2 also missed real spirals** — falsified by positive controls

Criterion	Run-level trips / 45,000	Stable points / 300	Why the count changed
v3 (windowed excess, normalized)	**0**	**300**	scale-windowed statistic is not streak-brittle; agent-normalized denominator prevents growth false positives — **and positive controls confirm it still catches real spirals**

v2 and v3 agree that the honest economy is stable (0 trips); they differ on whether *spirals* are caught, which the sweep cannot show (it contains no spirals) and only the positive controls can. That is precisely why the controls, not the sweep, are the load-bearing validation of v3.

How the floors were derived (so a reviewer can audit, not trust)

Floors are **not asserted**; they are the committed output of `sweep/derive_noise_floor.py` (artifact: `results/sweep_reports/noise_floor_derivation.json`). Method:

1. Run the honest economy across **all 300 sampled points** — 300 LHS points × 3 seeds × 3 demand variants (2,709 runs) — plus **growing-economy honest runs** (5–25 agents/epoch) so the noise model matches a launching exchange, plus the four controls.
2. For each run measure the peak E(W) at every (grace in {7,10,12,14}, window in {3,6,12}).
3. Set F(W) = **1.25 × max honest E(W)** (the safety factor: a 25% band above the worst honest run anywhere in the swept space, so an unlucky honest seed does not trip).
4. Keep only (grace, window) scales where F(W) sits **below** the weakest should-trip control with real margin. Report the margin; drop scales that overlap.

Separation table (grace 12, the operative floors):

Window	honest p50	honest p99	honest max	floor (1.25×)	weakest spiral (A/C/D)	negative ctl B	margin
W=6	0.171	0.315	0.368	**0.46**	0.524 (D)	0.171	**+0.064**
W=12	0.245	0.428	0.501	**0.63**	0.981 (D)	0.299	**+0.354**

W=6 detects fast (latency ~8 epochs) but its margin is thin (0.064) against the gentlest spiral (control D, ×1.18/epoch); W=12 is the wide-margin backstop for slower spirals. Below ~×1.055/epoch a spiral sits inside honest noise and is indistinguishable — the honest floor of detectability, stated plainly. The negative control (ring-farming) never exceeds 0.30 at any scale, well clear of both floors.

Positive & negative controls (validation in both directions)

`scenarios/controls_positive.py`, run on in-sample (42–44) and out-of-sample (100–102) seeds. Each control declares whether it *should* trip; certification requires every should-trip control to trip and every should-not control to stay silent.

Control	Design	Should trip	Result	Latency W6 / W12
A credit-line inflation	lines ×1.35/epoch, detector on	yes	(pass) trips (+ adversary_finding)	~8 / 14
B ring-farming	Sybil rings, wash detector DISABLED	**no**	(pass) silent (all seeds)	—
C camouflaged spiral	A + wash-padded volume, detector on	yes	(pass) trips (padding stripped)	8 / 14
D clean distributed spiral	×1.18/epoch under L_cap, detector on	yes	(pass) trips (**supply criterion alone**)	~8 / 14

Control B is *correctly* silent: balanced rings net to zero in credit outstanding (the Sybil cohort's aggregate negative balance is pinned all run), so ring-farming inflates volume, not credit stock — a fake-volume attack (the detector's domain, disabled here by construction), not a spiral. Under valid mutual-credit accounting a credit spiral is only reachable by inflating recorded lines outside settlement, which controls A/C/D exercise and which leaves two complementary signatures: the supply criterion (E(W)) and, past L_cap, a ledger-invariant violation. Control D isolates the supply criterion by staying under L_cap — it trips with **no** invariant backstop, proving the supply criterion itself carries the detection. (DECISIONS #31.)

Control E — the detector-DoS mirror, and the growth finding it surfaced

The wash-filtered denominator (#30) raises a mirror risk: an adversary who induces wash FALSE-POSITIVES against honest counterparties shrinks the qualified denominator, potentially inflating honest E(W) toward the floor (a denial-of-service on the honest exchange). `scenarios/control_e_detector_dos.py` measured it. Two results:

1. **The detector-DoS is a non-threat.** A *constant* induced-FP fraction cancels in the log-difference (Δlog of a constant-scaled series is unchanged), so it moves E(W) by 0. A *ramped* suppression adds < 0.01, because net qualified volume still grows faster than the ramp strips it. No EMA-damped denominator is required; damping would not even have helped.
2. **It surfaced a bigger bug the static-population sweep hid: growth-induced false positives.** A legitimately *growing* exchange (continuous registration, LS §4) false-tripped at **zero attack** — honest E(12) hit 0.66 (> the 0.63 floor) at 25 agents/epoch onboarding, because new agents draw credit lines before their settlement volume ramps, so credit-to-volume rises during onboarding and reads as a mild spiral. **Fix:** the active-agent term (DECISIONS #34). A real spiral inflates credit *per agent* (count flat → term 0 → still caught: controls A/D unchanged at E(12) ≈ 1.0); healthy growth inflates credit *with* the agent count (term cancels it: growth-25 honest E(12) 0.66 → 0.30). After the fix, Control E clears every floor under ramped suppression on a growing economy at every induced-FP rate tested (0–40%):

scenario	E(6)	E(12)	floors 0.46 / 0.63
ramp + growth 10/epoch	0.21	0.34	clears
ramp + growth 25/epoch	0.20	0.30	clears

   This is a concrete instance of why production floors cannot be inherited from simulation: the static-population noise model missed an entire class of honest behavior until a positive control forced it into view.

Coupled-subsystem CI lock

Because the criterion's denominator now depends on the wash detector, the floors depend on two subsystems: detector parameters and killcriteria code. `src/isonomia/calibration_lock.py` hashes (killcriteria source + detector config block); the derivation stamps that hash into its artifact; `tests/test_calibration_lock.py` **fails** if either changes without a matching re-derivation. Coupled subsystems re-calibrate together or the build breaks — they cannot silently drift. (DECISIONS #33.)

Production floors do not transfer from simulation — re-derive them on testnet

The floor VALUES here (grace 12, F(6)=0.46, F(12)=0.63) are simulation artifacts and must not be shipped to production. They were derived from this simulation's honest-noise distribution, which reflects this model's agent mix, demand process, capacity dynamics, and onboarding rate — none of which will match a live testnet exactly. Control E already demonstrated the failure mode: a noise model missing continuous-registration growth produced floors that false-tripped a growing economy, until growth was added to the sample. A different production reality (different growth curve, different default rate, different task-size distribution) will shift the honest E(W) distribution again.

The methodology transfers; the numbers don't. During the bootstrap grace window on testnet — while credits are still valueless and no halt has consequences — run `sweep/derive_noise_floor.py` against the *testnet's own* honest-epoch data (pass the testnet seeds), inspect the separation table against the same positive controls, and set the production floors from that. Then let the CI lock (above) hold the detector/criterion/floor triple together for the rest of the deployment. The simulation's contribution is the *shape* of the criterion (windowed excess, wash-filtered and agent-normalized denominator, the scales that separate, the 1.25× safety factor, the positive/negative control battery) — not the specific thresholds.

§8 parameter-registry recommendation (Sim Plan deliverable 5)

Stability does not discriminate within the swept ranges, so recommendations rest on the scenario evidence and mechanism findings. **Verdict: retain every §8 launch hypothesis.**

Symbol	Launch value	Recommendation	Evidence
D_erg	8 ergs	**Keep 8; treat ≥ 6.7 as a hard floor, ≥ 8 as the safe margin**	Supply-stability indifferent (flat terciles). S4: at every tested D_erg extraction = 0, but below 200/30 ≈ 6.7 the credit floor contracts and turnover farming achieved lines *above* bond — Sybil defense then leans on the wash detector rather than arithmetic. D_erg ≥ 8 keeps the collateralization invariant self-enforcing.
α	0.25	Keep 0.25	No instability observed among sampled values spanning 0.05–0.6; v1 noise mildly favored higher α (earned headroom smooths demand swings). 0.25–0.5 all defensible.
β	0.005	Keep 0.005	No instability observed among sampled values spanning 0.001–0.03. S6: fee-bleed attack uneconomic at 0.005 (capacity decay escapes it). Note: β prices *listed capacity*, and at no swept value does it counteract quality concentration — that is matching-driven, not a β dial.

Symbol	Launch value	Recommendation	Evidence
settlement fee	1.0%	Keep 1.0%	Converges to the ~0.7–0.9% cost-recovery band within 2–3 epochs from the 1.0% start; both sweep extremes (<0.5%, >1.7%) showed elevated v1 noise.
L_cap	10 × L_floor	Keep 10× (scaling L_floor_active, DECISIONS #12)	No instability observed among sampled values spanning 3–25×.
p*	0.50	Keep 0.50	No instability observed among sampled values spanning 0.35–0.65; basket calibration hits the target by construction and the epoch-6 retarget stays inside δ under attack (S2).
k	25	Keep 25	No stability or scenario signal across 5–100.
λ	2.0	Keep 2.0 — **flag as untested-in-anger**	No version transitions occur in Path A scope; λ never binds. Path B should not treat it as calibrated.
Kleos half-life	180 d	Keep 180 d	No instability observed among sampled values spanning 60–540 d; S7's patient-cluster share *decays* under it.
capacity_min	1 task/epoch	Keep 1	No instability observed among sampled values spanning 1–5.
Jury size / seed rate / duty quota	5 / 2% / 8	Keep	Not stressed beyond stubs (disputes stochastic per Sim Plan §1; seed rate fixed per §4).

Registry additions Path A recommends the spec adopt as named parameters: the v3 supply-criterion constants (grace 12, windows {6, 12}, floors {0.46, 0.63}, safety factor 1.25 — the floors flagged as re-derive-on-testnet per the section above); the wash-detector thresholds (bidirectionality requirement, conservation net-ratio 0.12 with top-3-counterparty share 0.70, trivial-spam share/count floors 0.5/8, robust-z 3.0 on median/MAD); the agent challenge threshold (10 review-upheld flags, LS §9 exclusion-pending-resolution); and Auditor review sensitivity (0.90 stub) — activation-clock integrity degrades roughly linearly in (1 - sensitivity), so it belongs in the registry, not in the implementation's shadows. The criterion now formally depends on the detector parameters (they define the qualified denominator), so the spec should note the two are a **coupled calibration unit** (DECISIONS #30, #33).

Attack-scenario verdicts at the recommended registry (unchanged from M5)

S1 wash rush: negative leakage in all three disclosure variants (full evasion -7.8%/attempt). S2 median drag: SCU drift 0.006 difficulty units, inside δ. S3 capacity flood: envelope binds, 94% overflow, victim income 0.97× twin. S4 Sybils:

extraction 0 at every D_erg (caveat above). S5 griefing: strictly positive attacker cost. S6 fee bleed: both policies escape via capacity decay. S7 patience: peak decisive power 28.8%, below the 1/3 blocking threshold, decaying.

Residual risks and honest caveats

- **Auditor accuracy is the soft underbelly**: wash-flag review at 0.90 sensitivity leaves ~10% of true-wash flags cleared; the challenge mechanism catches persistent rings anyway, but the leakage floor scales with review error. Not a parameter dial — an operations bar.
- **Quality stratification** is the honest economy's dominant structure: winner-take-most matching pins the bottom half at its credit floor (suspension guard holding). The credit-to-volume *level* (~3–3.5×) is structural; only its growth is pathological.
- **λ and jury machinery** ship uncalibrated by design (out of Path A scope).
- Kill criteria that never bind in-sweep cannot *rank* parameter values; the registry verdicts above are "no evidence against the hypotheses + scenario support", not optima.
- **The supply criterion detects spirals ≥ ~×1.055/epoch sustained.** Slower credit spirals sit inside honest noise and are indistinguishable by this statistic — the honest floor of detectability, not a tunable. W=6 detects fast spirals within ~8 epochs; W=12 catches slower ones later. A patient sub-threshold spiral is a residual the constitution should know about.
- **The v3 floors are a coupled calibration unit with the wash detector and must be re-derived on testnet** (see the production-floors section). The CI lock enforces this in-repo; it cannot enforce it across the sim→testnet boundary — that is a deployment discipline.

Go/no-go

Go. Path A's mandate — determine whether a stable operating region exists before any contract is written (Sim Plan §6) — is answered affirmatively for the sampled points: all 300 sampled Latin-hypercube points passed, forming one mutual-3-NN component within the sampled §4 domain, the seven scripted attacks fail against the calibrated defenses, and the supply kill-criterion is now validated in **both** directions (honest noise does not trip it; positive controls confirm real spirals do). The four defects Path A found were all in the constitution's own instrumentation — the §10 supply criterion — not in the economy:

1. v0 halted every launch at bootstrap → fixed (grace + log-convexity), **codified in Launch Spec v0.3 §10**.
2. v1 carried a ~5% false-positive rate on shock transients → fixed (magnitude floor).
3. v2 was blind to real spirals (streak brittleness) → fixed (windowed excess), **caught only because positive controls were demanded** — negative controls had passed it.
4. v3's wash-filtered denominator + a static-population noise model false-tripped growing economies → fixed (agent normalization + growth in the noise sample).

The through-line for Path B and the Launch Spec: **a kill criterion is itself a mechanism that must be adversarially tested.** The composite supply criterion, its positive/negative control battery, the auditable floor-derivation script, and the coupled-subsystem CI lock are as much a Path A deliverable as the economic parameter region. Nothing in the economy resisted validation; the instrumentation took four iterations — and would have shipped broken twice (v1 over-halting, v2 under-halting) without the positive controls and the growth stress test.